

SQL Server DBA, AWS SQL and AZURE SQL — Scenario based Real-World Questions

Performance / Tuning / Bottlenecks

- A query that used to run in 2 seconds is now taking 2 minutes — walk me through how you'd troubleshoot and resolve it.
- You see CPU usage at 100% on the SQL Server instance. How do you isolate the cause?
- Users complain of blocking / locking; multiple sessions are stuck. How do you find which queries/blockers and resolve the situation?
- You detect frequent deadlocks in your OLTP database. What steps do you take to analyze, resolve, and prevent future deadlocks?
- A stored procedure is consuming excessive tempdb space or causing spills. How will you diagnose and fix it?
- You notice high I/O wait times on the storage subsystem. How do you correlate SQL Server queries or operations to the underlying disk/subsystem?
- Indexes are heavily fragmented, and performance is impacted. What index maintenance plan would you implement (rebuild vs reorganize vs defragment)?
- You identify a missing index suggestion via the missing index DMVs. How do you evaluate whether to add it (cost/benefit)?
- Statistics are stale, causing poor execution plans. How do you detect that, and what strategy do you apply?
- Parameter sniffing is causing plan regression. What methods do you use to mitigate it?
- A large table is scanned frequently — how would you redesign (partitioning, filtered indexes, etc.) to optimize performance?
- Under high concurrency, you see latch contention on buffer pool or page latches. How do you track and fix it?
- A long-running transaction is blocking log truncation. How do you identify it and resolve without risking data integrity?
- You are asked to enforce SQL Server performance SLAs (latency, throughput). How do you monitor and alert?
- After deployment of a schema change, performance drops. What rollback or mitigation strategy do you follow?

Availability, HA, DR, Recovery

- A production server crashes unexpectedly. Walk me through your process to bring it back (restores, logs, business continuity).
- One node in an AlwaysOn Availability Group fails. How is failover handled? How do you bring it back?
- You need to set up a disaster recovery solution for minimal data loss. Which of the SQL Server high-availability/DR technologies do you choose, and why (AlwaysOn, log shipping, mirroring, replication)?
- A database backup fails with "backup media is corrupt." How do you recover from that?
- You discover that backups have not been running for a week, but no one noticed. How do you handle that, and how do you prevent recurrence?
- Transaction log backups are not truncating; the log file keeps growing. How do you debug that?
- You need to restore a database to a specific point in time (PIT). Walk me through your steps.
- The log file has grown extraordinarily large. How would you shrink it (if at all)? What are the risks?
- You want to take a database snapshot before doing a risky update. Under what conditions can you use database snapshots? What are their pros/cons?
- For a multi-region deployment, you need geo-distributed replication. How would you architect that for SQL Server?

Storage, Disk, Capacity Planning

- Disk space is low on the data drive. How do you figure out which databases/files are growing and free up space safely?
- The transaction log file has many VLFs (Virtual Log Files) and is causing performance issues. How do you reduce the number of VLFs?
- You have multiple data & log files across multiple disks. How do you distribute them for optimal I/O?
- You suspect I/O latency is causing slow queries. How do you validate, and what remediation steps do you take?
- You need to plan storage for a new large database (say several TBs). What factors do you consider (autogrowth, filegroups, striping, etc.)?
- A database file is set to grow automatically in small increments. Under heavy usage, it causes auto-grow events. How do you prevent or mitigate that?
- A backup file is enormous. How do you optimize backup storage (compression, backup devices, offloading to cloud)?
- You're requested to archive old data to reduce the main database size. How would you plan and implement that (partitioning, data purging, ETL)?

Security, Auditing, Compliance

34. You detect an unauthorized login or potential intrusion. What steps do you take to investigate and mitigate?
35. You need to ensure that certain columns (e.g. SSN, credit card) are always encrypted and not visible to DBAs — how would you enforce that?
36. You must audit all DML changes on a sensitive table. How would you set up auditing or change data capture / change tracking?
37. You are asked to enforce role-based access for developers, QA, production — how do you design grants and schema separation?
38. A developer wants to run ad-hoc SELECT queries in production on sensitive tables. They request read access. How do you handle that such that audit / security is preserved?
39. You need to enforce encryption in transit. What features do you enable (SSL/TLS, forcing encryption, certificates)?
40. You require that data at rest is encrypted. Which SQL Server encryption features can you use (TDE, Always Encrypted, cell-level encryption)?
41. You need to comply with GDPR / HIPAA. What database-level and server-level controls do you implement (masking, audit, data retention)?
42. A user reports that they lost access to a database. How would you recover from a dropped login or orphaned user issue?

Maintenance, Jobs, Automation

43. A SQL Server Agent job has failed nightly; alerts were not triggered. How do you detect and fix the missing alerting?
44. You have dozens of databases. How do you automate maintenance (indexing, statistics, integrity checks)?
45. You need to roll out schema changes to multiple environments (Dev / Test / Prod) with minimal downtime. How do you plan it?
46. A cross-database job needs to run backups, ETL, cleanup tasks. Describe how you schedule and manage dependencies.
47. You want to track schema changes over time (DDL history). How do you capture that?

48. You want to version-control stored procedures, functions, triggers, etc. What process or tools would you adopt?
49. You find that maintenance tasks (index rebuilds) are causing blocking during business hours. How do you adjust the schedule or algorithm?
50. You need to deploy new indexes on a large table with minimal downtime. What strategy do you use (online indexing, partition switching, etc.)?

Migration, Upgrade, Patching

51. You are asked to upgrade SQL Server from 2014 to 2022. How would you plan for compatibility, downtime, rollbacks?
52. There are deprecated features in your database code when moving to a newer SQL Server version. How do you identify and remediate them?
53. You have a legacy database and need to migrate it to a new instance with minimal downtime. What are migration strategies?
54. You choose to move from on-prem to SQL in a cloud VM. What architectural changes (e.g. I/O, storage, networking) do you need to account for?
55. You need to patch SQL Server with security updates. How do you apply updates to availability groups while preserving high availability?
56. You need to migrate from one server to another without downtime. How do you replicate data live (log shipping, replication, etc.) and switch over?

Troubleshooting, Incident Handling

57. A database is in suspect mode. How do you recover it?
58. You find corruption on a data page. What tools and steps do you use to diagnose and possibly repair it (DBCC CHECKDB, DBCC PAGE, backups)?
59. An index rebuild is stuck or blocking other operations. How do you intervene?
60. A query plan is missing a needed filter. The developer claims their query is correct. How do you analyze and coach them?
61. A sudden massive spike in writes causes the log disk to saturate and slow everything. How do you control, throttle, or mitigate until you fix?
62. A catastrophic hardware failure occurred for a server. Walk me through how you bring up your DR environment, recover databases, point applications to it.
63. A schema change fails halfway (e.g. adds a column and updates millions of rows). How do you resume or rollback safely?
64. You see that the log shipping latency is high — secondary server is lagging. How do you find root cause and reduce delay?
65. An AlwaysOn secondary node is out of sync. How do you re-synchronize it without losing data?

Azure SQL / Azure SQL Database / Azure SQL Managed Instance — Scenario Questions:

1. Your Azure SQL Database is experiencing high CPU and DTU/vCore saturation during peak hours. How will you diagnose and mitigate?
2. After deploying a change, database performance degraded. How would you use Query Store or Performance Insight to identify regressed queries?
3. Users are reporting occasional connection timeouts. How would you troubleshoot from the SQL and networking side?

4. You need to migrate an on-premises SQL Server to Azure SQL Database, with minimal downtime. What tools and strategies do you use?
5. You want geo-redundancy and ability to fail over across regions. How would you configure it?
6. Your Azure SQL cost is escalating. How do you analyze, optimize, and right-size your deployment?
7. A business requires near-zero downtime for maintenance windows. Which Azure SQL features would you use (e.g. auto-failover groups)?
8. You detect unauthorized access attempts. How do you investigate and remediate using Azure's security features?
9. You must encrypt sensitive data fields so that even DBAs can't see them. Which features would you use (Always Encrypted)?
10. You need to audit all data changes on specific tables. How would you set that up in Azure SQL?
11. Database size has hit the max storage limit. What strategies do you employ (scaling, archiving, sharding)?
12. You want to automate deployment of Azure SQL databases (schema, settings, firewall) — how would you use ARM templates / Terraform / Bicep?
13. You want to integrate with Azure Key Vault for certificate and encryption management. How would you architect that?
14. You find that the Azure SQL database is paused (Serverless mode) and queries are slow to resume. How do you manage or mitigate that?
15. You need to configure high availability within the same region. What HA capabilities are available (e.g. zone redundant, business critical)?
16. You notice that index maintenance tasks are unavailable in Azure SQL (because of PaaS limitations). How would you work around that?
17. You want to scale multiple databases as load changes. How would you use Elastic Pools to optimize cost?
18. You have multiple databases that share similar workloads. How would you consolidate or manage them more efficiently?
19. You need cross-database queries in a managed instance. What constraints and solutions exist?
20. Your application occasionally executes long-running transactions that block other workloads. How would you manage concurrency / isolation / tuning in Azure SQL?
21. Your application is multi-tenant. Do you choose one large database or multiple tenant-specific databases in Azure SQL? What are tradeoffs?
22. You detect a regression in query performance after statistics update. How do you identify and force a stable plan?
23. You need to enforce IP firewall rules, virtual network rules, and use Private Link / service endpoints. How do you secure connectivity?
24. You want to shift from unmanaged VM-based SQL to Azure SQL Managed Instance. How would you plan and execute the migration?
25. During a regional Azure outage, your database should fail over to a secondary region. How will you test and measure RPO & RTO?
26. You need to integrate with Azure Active Directory authentication for database logins. How do you configure that securely?
27. A query is using a lot of memory grant and is being throttled. How do you diagnose memory grant issues in Azure SQL?
28. You find TempDB contention in the managed instance. How would you address it?
29. You want to shift to a serverless tier to save cost in off-hours. What tradeoffs do you evaluate?
30. You suspect a query plan regression due to parameter sniffing. How do you mitigate in Azure SQL?
31. You have to migrate data from AWS RDS SQL Server to Azure SQL. What steps would you take?
32. You want to automate backups retention, long-term backups offloading to Blob storage — how do you set that up?
33. A critical patch is released. How do you ensure minimal downtime patching across multiple Azure SQL instances?

34. You want to monitor performance, define alerts, KPIs (e.g. DTU %, CPU, I/O). How do you set up monitoring (Azure Monitor, SQL Insights)?
35. You need to clone or snapshot a database for testing environment in Azure. What features do you use?
36. The application team complains that some queries are slower than on-prem. What differences in Azure SQL architecture do you consider (network latency, storage, caching)?
37. You deploy a complex schema change involving large table rebuild. How do you plan it in Azure SQL with limited downtime?
38. You want to ensure business continuity in region failover. How is failover group / auto-failover designed?
39. You need to test disaster recovery. How do you simulate failover and validate application correctness?
40. You are in charge of optimizing data ingress/egress costs (bandwidth, data movement). What strategies do you use to minimize cloud egress expense?

AWS SQL (e.g. Amazon RDS for SQL Server, Aurora, etc.) — Scenario Questions

1. You host SQL Server in Amazon RDS. CPU and IO are saturated. Which metrics and tools (CloudWatch, Enhanced Monitoring, Performance Insights) would you use to diagnose?
2. RDS instance is running out of storage. How do you expand storage? What are implications for IOPS, performance, downtime?
3. You need a read replica to offload reporting traffic. How do you set up replication in RDS for SQL Server?
4. You want high availability across AZs. How do you configure Multi-AZ in RDS?
5. You need to run customizing SQL Agent jobs / scheduled tasks in RDS (which is managed). How do you handle that constraint?
6. Your application team complains about slow cross-region queries. How do you optimize, possibly using read replicas or caching?
7. You detect a sudden spike in connection count and many failed logins. How do you trace that?
8. During a failover event, how do you ensure minimal downtime for RDS?
9. You want to encrypt data at rest and in transit. How do you enable encryption in RDS SQL Server and enforce TLS?
10. You must migrate from on-prem SQL to RDS with minimal downtime. What migration strategies would you use?
11. You want to automate backups & retention. How do you configure backup windows, snapshot lifecycles, export to S3?
12. You see increased latency due to network in cloud. How do you trace effects of VPC, subnets, security groups, routing?
13. The RDS instance's maintenance window is upcoming. You need to apply patches. How do you schedule and test updates while ensuring high availability?
14. You want to offload analytics workloads to a read-only instance. How do you isolate and manage workloads?
15. A cross-region DR requirement arises. How do you replicate your RDS SQL Server to another region?
16. You notice storage throughput throttling due to bursting credits or IOPS limits. How do you monitor and correct?
17. Your log backups are failing due to permission issues. How do you fix that in a managed environment?
18. Developers need access to run queries on production occasionally. How do you provide limited access while preserving security?
19. You have a requirement to scale out read workload dynamically. How do you design with RDS and perhaps caching / ElasticCache?
20. You want to monitor slow queries in RDS. How do you enable query logging, slow query insight, and how do you act on that information?

Mixed / Variations / “Twist” Scenarios & Edge Cases

1. A sudden schema change causes data corruption in production. You cannot roll back easily. How do you mitigate and correct?
2. A bug causes massive data deletion. You have to recover lost rows from backups and logs. How do you approach it?
3. There is a network partition between app servers and DB servers for 15 minutes during peak traffic. How do you design to tolerate it?
4. The business wants to scale beyond what a single database instance can support (e.g. 100 TB). How do you design a sharded / distributed database architecture?
5. You have to migrate a live system across cloud providers (e.g. AWS → Azure). How do you plan and execute?
6. The application issues unparameterized queries resulting in SQL injection vulnerabilities. How do you detect, fix, and future-proof?
7. The app must support multi-tenant data isolation. How do you implement row-level security / schemas / database per tenant?
8. The developers want full control to change database schema. How do you create a process to validate, audit, and deploy such changes safely?
9. You are under pressure from management: “We need zero downtime for migration.” What compromises or strategies will you propose?
10. You find that two critical systems share the same SQL Server but cause performance interference. How do you isolate resources? (Resource Governor, separate instances, etc.)
11. You inherit a database with zero documentation and no monitoring. How do you capture knowledge, baseline, and manage forward?
12. An audit reveals unauthorized direct table access via SSMS. How do you tighten the system, and how do you retroactively detect what was accessed?
13. You are tasked with designing a self-healing / auto-scaling database architecture. What automation, alerting, scaling policies do you include?
14. The app uses heavy reporting workloads which slow down OLTP. How do you offload, isolate or replicate to minimize interference?
15. A change in business logic causes a sudden surge in writes (x2). The database is unable to cope. What do you do in short and long term?
16. Your cloud SQL instance is nearing its maximum IOPS/throughput. How do you architect to continue scaling?
17. The data model (ER schema) has serious design flaws (many-to-many bridging, missing normalization). How do you refactor in production with minimal disruption?
18. You want to run large ETL jobs during off-peak, but they interfere with business queries. How would you schedule / throttle / isolate?
19. You must support real-time analytics on transactional data without locking the primary database. How do you design such a pipeline?
20. You need a mechanism for zero-downtime schema changes, like adding columns or indexes, on large tables. What patterns do you use?

Section A: SQL Server DBA (On-Premises) – 150 Questions

1. Your SQL Server full backup is 500 GB, but the differential backup is 600 GB. What could be the reason?
2. A user complains that their query is running slow today but was fine yesterday. How would you troubleshoot?
3. The transaction log file has grown to 500 GB. What steps will you take to reduce it safely?
4. You see blocking in SQL Server. How do you identify the blocking session and resolve it without killing production sessions?
5. How do you recover a database that went into suspect mode?
6. How do you migrate a SQL Server database from one server to another with minimal downtime?
7. You have 10 SQL Agent jobs failing. What steps do you take to troubleshoot and fix?
8. Users report “Login failed for user” intermittently. How do you debug?
9. Database mirroring has stopped synchronizing. What steps do you take?
10. The AlwaysOn Availability Group is in Resolving state. What is your plan of action?
11. How would you patch a production SQL Server with AlwaysOn configured?
12. TempDB keeps running out of space. What real-time fixes do you apply?
13. A query is running fine in QA but extremely slow in Production. How would you troubleshoot?
14. What would you do if your primary database data file is corrupted but log file is intact?
15. Your DB backups are completing, but restore testing shows corruption. How do you proceed?
16. You notice excessive deadlocks in production. How do you reduce them?
17. The error log shows frequent auto-growth events. How would you fix it?
18. A database is in “Restoring” state after a failed restore. How do you fix it?
19. How do you recover from a dropped table without recent backups?
20. A SQL Server cluster node failed. What’s your next step?
21. How do you handle a failing index rebuild job that times out?
22. A critical stored procedure suddenly starts consuming 90% CPU. What do you do?
23. The SQL Server service won’t start. How do you troubleshoot?
24. You see IO bottlenecks on data drives. How would you diagnose and fix?
25. How do you perform root cause analysis for a SQL Server crash?
26. DBCC CHECKDB reports corruption. What recovery options do you consider?
27. Users complain of high login latency. What would you check?
28. How do you migrate logins and jobs during a SQL Server migration?
29. A database restore is taking much longer than expected. What can you do?
30. How do you identify the largest tables consuming storage?
31. How would you encrypt backups for compliance?
32. How do you troubleshoot high PAGEIOLATCH waits?
33. Your SQL Server agent service won’t start. What’s your fix?
34. A developer dropped an index accidentally. How do you recover quickly?
35. AlwaysOn secondary replica is not synchronizing. How do you fix?
36. Your maintenance job shrinks logs every day. Is this best practice?
37. SQL Server memory keeps increasing without releasing. How do you handle?
38. The system database (msdb) is corrupted. What’s your recovery plan?
39. A large SSIS package migration failed. How do you troubleshoot?
40. A job running a long query blocks other transactions. What do you do?
41. A login keeps getting orphaned after restore. How do you fix?
42. Backups are consuming too much storage. How do you optimize?
43. A SQL Server upgrade fails halfway. How do you recover?
44. Users complain of intermittent timeouts. Where do you start troubleshooting?
45. SQL Profiler shows parameter sniffing issues. How do you fix?
46. What’s your approach for index fragmentation above 80%?

47. A query plan suddenly changed causing slowness. What would you do?
48. How do you move TempDB to another drive?
49. What steps do you take to harden SQL Server security?
50. Your database has thousands of VLFs. How do you fix it?
51. A transaction log backup chain is broken. What's your recovery?
52. How do you implement log shipping without downtime?
53. SQL Server licensing expired. What happens to databases?
54. How do you troubleshoot high SOS_SCHEDULER_YIELD waits?
55. SQL jobs run fine manually but fail on schedule. Why?
56. Your database grows rapidly. How do you monitor growth trends?
57. How do you clone a production database for testing?
58. How do you fix orphaned users after a restore?
59. Database snapshots consume unexpected space. Why?
60. A developer requests sysadmin rights. How do you handle?
61. SQL error log is 20GB. How do you manage it?
62. The model database is altered. What impact does it have?
63. Your differential backup size is equal to a full backup daily. Why?
64. How do you detect implicit conversions slowing queries?
65. How do you patch SQL Server cluster nodes?
66. A replication snapshot agent fails. What do you do?
67. Latency in transactional replication increases. How do you troubleshoot?
68. Merge replication keeps reinitializing. Why?
69. How do you reconfigure replication after failover?
70. A publisher database is dropped accidentally. How do you recover replication?
71. Replication distribution database grows large. How do you fix?
72. Subscriber latency is 30 mins. How do you reduce it?
73. A peer-to-peer replication conflict occurs. How do you resolve?
74. SQL Agent alerts stop sending emails. How do you fix?
75. Extended events session grows out of control. How do you manage?
76. Performance counters disappear. What do you check?
77. DMV shows ghost records. How do you clean them?
78. Your index maintenance runs during peak hours. How do you optimize?
79. How do you troubleshoot latch contention?
80. SQL Server is experiencing spinlock contention. How do you diagnose?
81. What do you do if sys.dm_exec_requests shows many CXPACKET waits?
82. A temp table is causing recompilations. How do you fix?
83. You need to implement transparent data encryption. What's the process?
84. How do you implement auditing without major performance impact?
85. SQL Server login audit shows brute force attempts. How do you secure?
86. Database mail fails to send. How do you debug?
87. A CLR assembly fails to load. What do you check?
88. A job to purge old data fails due to lock escalation. How do you solve?
89. DMV shows high signal waits. What does it mean?
90. A SQL dump file is generated. How do you analyze?
91. Your SQL Server startup is slow. How do you investigate?
92. How do you troubleshoot excessive recompilations?
93. SQL memory clerk shows MEMORYCLERK_SQLBUFFERPOOL consuming 95%. What next?
94. Database mirroring witness is offline. What impact?
95. A transaction rollback takes hours. How do you handle?
96. You have orphaned replication subscriptions. How do you clean?

97. Log shipping secondary database is out of sync. What's your fix?
98. A job step using PowerShell fails. How do you debug?
99. Query execution plans differ between servers. Why?
100. CPU spikes happen daily at 2 AM. How do you find the cause?
101. SQL Server stops responding to new connections. What do you check?
102. You need to capture long-running queries. How do you configure?
103. A SQL Server login is disabled automatically. Why?
104. Application gets "Could not allocate space for object in database." What's wrong?
105. TempDB log file fills quickly. How do you monitor?
106. A developer created a recursive CTE that hangs. How do you fix?
107. A heap table grows rapidly. How do you optimize?
108. Partitioned table queries are slow. How do you troubleshoot?
109. Statistics updates cause blocking. How do you mitigate?
110. SQL Server Agent alert does not trigger. What do you check?
111. Blocking occurs due to schema lock. How do you fix?
112. Read committed snapshot isolation is enabled. What impact can it have?
113. How do you identify top queries consuming CPU?
114. Users complain of random disconnects. What's your root cause analysis?
115. Mirroring role change failed. What do you do?
116. SSRS reports fail due to login timeout. How do you troubleshoot?
117. A cluster witness goes offline. What happens?
118. SQL Server fails after OS patching. How do you roll back?
119. A page restore fails. What's your next step?
120. Query store is consuming too much space. What do you do?
121. You must migrate 10TB data overnight. How do you plan?
122. An audit requires login history. How do you fetch it?
123. Users complain of "string or binary data would be truncated." How do you resolve?
124. A DMV shows missing indexes suggestion. How do you validate before creating?
125. DBCC CHECKDB on a large DB takes too long. How do you handle?
126. How do you configure backup compression by default?
127. TempDB contention occurs on allocation pages. How do you fix?
128. A developer executes sp_configure without permission. How is it possible?
129. Database CPU usage is high due to parallelism. How do you tune?
130. SQL Server upgrade changes compatibility level. How do you handle?
131. How do you restore a system database (master)?
132. What do you do if the SQL Server agent msdb is corrupted?
133. How do you perform a tail log backup?
134. SQL Server shows out-of-memory errors. What do you check?
135. A clustered index rebuild takes hours. How do you optimize?
136. Query plans show key lookups. How do you reduce them?
137. Log shipping copy job fails. How do you troubleshoot?
138. A log backup job overlaps with a full backup job. What happens?
139. SQL Agent job history truncates. How do you extend it?
140. AlwaysOn listener fails to connect. What do you check?
141. Database stuck in restoring forever. Why?
142. Error log shows "Could not write to tempdb." What's your fix?
143. SQL Server shows excessive worker threads. How do you fix?
144. How do you reduce excessive recompilation due to temp tables?
145. Statistics are outdated but auto-update is ON. Why?
146. How do you detect parameter sniffing?

- 147. Backup jobs succeed but restore validation fails. Why?
- 148. Database mirroring role switch takes too long. How do you optimize?
- 149. SQL Server startup account password expired. What happens?
- 150. Query execution shows excessive spills to TempDB. How do you fix?

Section B: AWS RDS for SQL Server – 120 Questions

- 151. How do you enable SQL Server Agent jobs in RDS?
- 152. You need to restore a .bak file from on-prem SQL Server to RDS. How do you do it?
- 153. RDS instance is running out of storage. How do you scale without downtime?
- 154. How do you troubleshoot RDS failover events seen in CloudWatch logs?
- 155. A developer needs to run cross-database queries in RDS. What challenges will you face?
- 156. How do you configure automated backups & point-in-time restore in RDS?
- 157. Your RDS instance CPU is consistently 90%. What are your immediate steps?
- 158. How do you enable Transparent Data Encryption (TDE) in RDS SQL Server?
- 159. Your RDS instance was rebooted automatically. How do you investigate the root cause?
- 160. Can you configure linked servers in RDS? If yes, how?
- 161. How do you set up read replicas in RDS for SQL Server?
- 162. What's your approach to apply SQL patches in RDS?
- 163. You need to migrate 1TB SQL DB from on-prem to RDS. How do you plan?
- 164. How do you configure CloudWatch alarms for RDS performance?
- 165. RDS backups are failing. Where do you check first?
- 166. Multi-AZ RDS is failing over frequently. What's your troubleshooting path?
- 167. How do you secure RDS access from public internet?
- 168. A developer requests sysadmin access in RDS. How do you handle?
- 169. RDS automated backups are disabled. What impact does it have?
- 170. How do you reduce RDS storage costs?
- 171. A query execution time doubled after migration to RDS. How do you analyze?
- 172. How do you enable SQL Audit in RDS?
- 173. A restore to a point-in-time fails. What's your fix?
- 174. You need to copy RDS snapshots to another region. How?
- 175. RDS transaction logs are filling quickly. What do you do?
- 176. How do you test DR with RDS Multi-AZ?
- 177. A developer wants CLR in RDS. What do you say?
- 178. How do you upgrade RDS SQL Server engine version?
- 179. RDS parameter group modification failed. How do you fix?
- 180. How do you migrate logins to RDS?
- 181. How do you monitor blocked queries in RDS?
- 182. A client needs SQL profiler in RDS. What's the workaround?
- 183. RDS storage auto-scaling fails. Why?
- 184. How do you reduce downtime when scaling RDS instance class?
- 185. A cross-region replica is lagging. How do you fix?
- 186. How do you configure IAM authentication for RDS SQL Server?
- 187. How do you manage SSL/TLS certificates for RDS connections?
- 188. RDS upgrade fails. How do you rollback?
- 189. RDS free storage alerts are ignored. What could happen?
- 190. A query causes excessive tempdb usage. How do you manage in RDS?
- 191. RDS backup retention is set to 1 day accidentally. How do you recover?
- 192. How do you export RDS data to S3 securely?
- 193. RDS performance insights show waits. How do you use it for tuning?

194. Can you schedule RDS backups at specific times?
195. A restore requires renaming DBs in RDS. How do you do it?
196. RDS Multi-AZ failover took 10 mins. Why?
197. How do you configure connection pooling for RDS?
198. What happens if RDS backup window overlaps with maintenance window?
199. You must encrypt RDS snapshots. How do you implement?
200. An RDS instance hit max connections. What's your fix?
201. You need cross-account RDS snapshot sharing. How?
202. How do you configure RDS SQL alerting to PagerDuty?
203. RDS reports insufficient memory. How do you diagnose?
204. Multi-AZ replica lags behind primary. What do you do?
205. How do you restore an RDS DB to a different subnet group?
206. RDS option group update fails. Why?
207. How do you troubleshoot RDS high IOPS usage?
208. CloudWatch shows high write latency on RDS. What's your fix?
209. How do you reduce downtime during RDS parameter group changes?
210. A client needs FILESTREAM in RDS. Is it possible?
211. How do you archive logs from RDS to S3?
212. How do you configure RDS event notifications?
213. How do you estimate storage needs for RDS growth?
214. RDS log exports to CloudWatch fail. Why?
215. A query causes excessive deadlocks. How do you fix in RDS?
216. How do you enable RDS encryption after DB creation?
217. RDS instance restore is stuck. How do you resolve?
218. How do you handle GDPR compliance in RDS?
219. What happens if RDS primary AZ fails?
220. How do you tune RDS for bulk inserts?
221. How do you schedule RDS scaling during peak load?
222. CloudTrail shows unauthorized RDS API calls. What's your response?
223. How do you configure performance baselines for RDS?
224. RDS upgrade introduces compatibility issues. What next?
225. RDS DB parameter modification requires reboot. How do you plan downtime?
226. RDS CPU spikes every day. How do you diagnose?
227. A backup restore exceeds allocated storage. How do you fix?
228. How do you monitor blocking sessions in RDS without Profiler?
229. How do you ensure encryption at rest in RDS?
230. Can you use SQL Server Agent alerts in RDS?
231. What happens when RDS storage is 100% full?
232. How do you migrate RDS to Aurora?
233. How do you enforce SSL connections in RDS?
234. How do you test RDS failover time?
235. What happens if you delete RDS parameter groups?
236. How do you integrate RDS with AWS Secrets Manager?
237. How do you scale RDS IO throughput?
238. RDS restore to different engine version fails. Why?
239. How do you refresh lower envs with prod data in RDS?
240. Can you configure custom trace flags in RDS?
241. How do you enforce TLS 1.2 in RDS SQL?
242. A snapshot copy across regions is too slow. How do you optimize?
243. How do you export schema only from RDS?

244. How do you integrate RDS audit logs with SIEM?
245. RDS instance rebooted during backup. What happens?
246. Can you rebuild indexes in RDS?
247. RDS DB parameter modification causes downtime. How do you handle?
248. What's the impact of disabling automated backups in RDS?
249. How do you monitor RDS freeable memory?
250. How do you troubleshoot high connections in RDS?
251. How do you configure hybrid connectivity from on-prem SQL to RDS?
252. RDS SQL Server version upgrade caused breaking changes. What next?
253. How do you handle DB corruption in RDS without direct access to DBCC CHECKDB repair options?
254. RDS restore to a point-in-time DB is inconsistent. How do you validate?
255. You must migrate SQL Agent jobs to RDS. What's your approach?
256. RDS disk I/O throughput is throttled. What's your fix?
257. How do you monitor RDS query performance in real-time?
258. A client wants cross-region HA for RDS SQL. How do you design it?
259. How do you handle accidental RDS DB deletion?
260. You see high tempdb usage in RDS. What are your options?
261. How do you reduce RDS DB maintenance window downtime?
262. How do you enable CDC (Change Data Capture) in RDS?
263. How do you perform log shipping in RDS?
264. An RDS SQL instance is nearing max database limit. What's your fix?
265. You need to move RDS logs to S3 daily. How?
266. RDS SQL replication breaks after failover. How do you recover?
267. RDS DB encryption key is deleted. What happens?
268. How do you control RDS DB connections per user?
269. A query execution plan is missing in RDS. How do you get it?
270. CloudWatch shows "Burst Balance" depletion. What's the impact?

Section C: AWS EC2 with SQL Server – 110 Questions

271. Your SQL Server AlwaysOn on EC2 is showing high latency between AZs. How do you troubleshoot?
272. What's your process to apply SQL Server cumulative updates on EC2 with clustering?
273. An EC2 SQL Server crashed. How do you restore services quickly?
274. Your EC2 instance disk is full. What's your immediate recovery plan?
275. How do you configure CloudWatch alerts for SQL Server logs on EC2?
276. A SQL Server in EC2 must connect to on-prem via VPN. Latency is high. How do you optimize?
277. How do you perform encrypted backups from SQL Server EC2 to S3?
278. How do you configure AlwaysOn across two EC2 instances in different regions?
279. How do you recover from accidental deletion of EC2 SQL Server instance?
280. How do you secure EC2 SQL Server from public internet exposure?
281. How do you handle EC2 SQL patching with minimum downtime?
282. You need to expand EC2 EBS volume for SQL data. What's the process?
283. A SQL Server job in EC2 fails due to missing credentials. How do you fix?
284. EC2 SQL Server logs show storage throttling. What's your approach?
285. How do you configure SQL Server AlwaysOn Listener in AWS EC2 with Route53?
286. EC2 SQL Server backup to S3 fails intermittently. How do you fix?
287. EC2 instance reboot caused SQL crash. How do you diagnose?
288. How do you automate SQL patching on EC2 servers?
289. SQL Server on EC2 shows network packet drops. How do you fix?
290. EC2 SQL Server performance drops after migration from RDS. Why?

291. How do you integrate CloudTrail auditing for SQL on EC2?
292. How do you ensure high availability for SQL Server on EC2 without RDS?
293. EC2 SQL Server shows clock drift errors. How do you handle?
294. How do you design DR for EC2 SQL Server across regions?
295. SQL Server backups from EC2 to Glacier are failing. What's your fix?
296. EC2 SQL Server cluster fails after AMI update. What next?
297. You must migrate 5 TB database from EC2 SQL to RDS. How do you plan?
298. EC2 SQL Server gets brute-force login attempts. How do you protect?
299. SQL Server AlwaysOn sync between EC2 nodes is slow. How do you tune?
300. How do you configure IAM roles for EC2 SQL backups to S3?
301. How do you analyze EC2 SQL performance using PerfMon remotely?
302. Your EC2 SQL data disk performance drops. How do you check EBS settings?
303. EC2 SQL backup jobs fail after volume encryption. Why?
304. EC2 SQL server licensing expired. What's your fix?
305. How do you migrate SQL Server logins between EC2 instances?
306. AlwaysOn failover fails due to quorum issue in EC2 cluster. What's your step?
307. A database restore from S3 to EC2 SQL fails. How do you debug?
308. EC2 SQL CPU usage spikes after antivirus update. How do you mitigate?
309. How do you configure monitoring dashboards for EC2 SQL using CloudWatch + Grafana?
310. How do you rebuild an EC2 SQL cluster node after corruption?
311. SQL Server startup fails after Windows patching in EC2. How do you recover?
312. How do you automate EC2 SQL backups to cross-region S3?
313. EC2 SQL Server disk queue length is high. What do you check?
314. How do you perform blue-green deployment for SQL Server on EC2?
315. EC2 SQL AlwaysOn listener doesn't resolve DNS. How do you fix?
316. You must migrate SSRS/SSIS from EC2 to RDS. How do you handle?
317. EC2 SQL transaction log backups fail during high load. What's your solution?
318. EC2 SQL data disk corruption occurs. How do you recover?
319. How do you set up EC2 SQL patch management at scale?
320. SQL Server on EC2 shows licensing mismatch after AMI restore. Why?
321. How do you configure EC2 SQL with hybrid Active Directory?
322. How do you move 100 DBs from EC2 SQL to Azure SQL?
323. EC2 SQL memory pressure occurs. What tuning options exist?
324. A SQL Server on EC2 experiences failover cluster validation error. How do you resolve?
325. SQL backups on EC2 exceed S3 bucket quota. What's your fix?
326. How do you manage secrets/connection strings for EC2 SQL securely?
327. A query works in on-prem SQL but fails in EC2 SQL. Why?
328. How do you encrypt EC2 SQL data at rest and in transit?
329. EC2 SQL upgrade requires downtime. How do you reduce impact?
330. AlwaysOn cluster IP fails to register in EC2 VPC. How do you fix?
331. SQL Agent jobs disappear after migration. Why?
332. EC2 SQL is not accessible after firewall update. What next?
333. SQL Server cluster witness in EC2 fails. How do you recover?
334. EC2 SQL error logs show insufficient permissions. How do you fix?
335. How do you analyze EC2 SQL dumps with AWS tools?
336. How do you build automated DR drill tests for EC2 SQL AlwaysOn?
337. A snapshot restore of EC2 SQL corrupts data. What next?
338. How do you migrate SQL log shipping to EC2 setup?
339. How do you set up SQL patch rollback on EC2?
340. EC2 SQL RDP access is lost. How do you manage DB access?

341. SQL backups in EC2 take too long. How do you optimize?
342. How do you deploy SQL Server AlwaysOn in EC2 with Terraform?
343. EC2 SQL stops accepting connections after scaling. Why?
344. How do you simulate failover for SQL EC2 cluster?
345. EC2 SQL instance migration fails due to OS mismatch. What's your fix?
346. SQL Server crash dumps are not generated in EC2. How do you fix?
347. How do you integrate CloudWatch metrics with SQL DMVs for monitoring EC2 SQL?
348. EC2 SQL server instance runs into storage IO credits limit. How do you fix?
349. EC2 SQL migration to AWS Outposts fails. Why?
350. How do you configure hybrid log shipping from EC2 SQL to on-prem?
351. SQL Server licensing compliance check fails in EC2. What's your fix?
352. EC2 SQL SSRS service won't start. How do you debug?
353. AlwaysOn automatic seeding fails in EC2. How do you fix?
354. How do you automate monthly SQL patching in 50+ EC2 servers?
355. SQL backups from EC2 to cross-account S3 fail. What do you do?
356. A ransomware attack encrypts EC2 SQL files. How do you recover?
357. EC2 SQL shows Kerberos authentication failures. What's your fix?
358. SQL Agent job uses xp_cmdshell in EC2. Is it secure?
359. How do you configure DMS migration from EC2 SQL to Aurora?
360. EC2 SQL AlwaysOn witness fails in a different AZ. What's your step?
361. How do you implement TLS rotation in EC2 SQL?
362. EC2 SQL audit logs must be centralized. How do you configure?
363. A query plan cache is flushed daily. Why in EC2 SQL?
364. How do you build immutable backups for EC2 SQL?
365. EC2 SQL security group misconfig blocks replication. How do you fix?
366. SQL Server AlwaysOn failover causes DNS stale records. How do you fix?
367. EC2 SQL takes long to reboot after failover. Why?
368. How do you use AWS SSM to automate SQL admin tasks in EC2?
369. EC2 SQL server max worker threads exhausted. How do you tune?
370. How do you perform in-place upgrade for EC2 SQL with rollback?
371. EC2 SQL database mail fails. How do you debug?
372. A snapshot of EC2 SQL fails due to VSS error. What's your fix?
373. SQL EC2 clustering fails with multi-subnet setup. How do you fix?
374. How do you implement monitoring for SQL AlwaysOn in EC2?
375. SQL Server in EC2 goes into recovery pending. How do you recover?
376. SQL EC2 CPU goes 100% during index rebuild. What's your plan?
377. A DB file growth fails due to EBS limit. How do you resolve?
378. SQL EC2 connection pooling causes timeouts. How do you tune?
379. EC2 SQL audit shows privilege escalation. What's your response?
380. How do you automate SQL deployment scripts across EC2 fleet?

Section D: Azure SQL Database & Managed Instance – 120 Questions

381. A query runs in 2 sec in on-prem SQL but 45 sec in Azure SQL. How do you troubleshoot?
382. How do you implement failover groups for Azure SQL databases?
383. Your Azure SQL database has reached max DTU limit. What's your approach?
384. How do you restore a deleted Azure SQL database?
385. A user needs cross-database joins in Azure SQL. What options are available?
386. Your Azure SQL database shows many throttling events. How do you resolve?
387. How do you migrate a 5 TB on-prem SQL DB to Azure SQL Managed Instance?

388. How do you enable auditing & threat detection in Azure SQL?
389. How do you configure geo-replication for disaster recovery?
390. Your backup retention needs to be 10 years. How do you configure in Azure SQL?
391. Query store grows too large in Azure SQL. How do you manage?
392. How do you tune a query that uses excessive DTUs in Azure SQL?
393. Azure SQL logins don't sync with AAD. How do you fix?
394. How do you migrate SQL Agent jobs to Azure SQL?
395. Azure SQL automatic tuning changed indexes incorrectly. What's your fix?
396. You must configure private endpoint access for Azure SQL. How?
397. How do you restore Azure SQL to a different subscription?
398. Azure SQL long-running query kills other sessions. What do you do?
399. How do you enforce TLS 1.2 in Azure SQL connections?
400. Azure SQL firewall blocks app traffic. How do you fix?
401. How do you configure point-in-time restore in Azure SQL?
402. Azure SQL connection pool exhausts. How do you tune?
403. How do you monitor query performance in Azure SQL using Query Insights?
404. A user requests sysadmin role in Azure SQL. How do you handle?
405. How do you automate scaling for Azure SQL elastic pools?
406. How do you configure failover between two Azure regions?
407. A schema drift occurred after migration. How do you fix?
408. How do you integrate Azure SQL auditing with SIEM tools?
409. A developer needs xp_cmdshell in Azure SQL. What's your response?
410. Azure SQL DTU usage spikes randomly. How do you investigate?
411. How do you enforce encryption at rest in Azure SQL?
412. Azure SQL backup restore fails. What's your fix?
413. Azure SQL connection latency increases. How do you tune?
414. How do you configure high availability in Azure Managed Instance?
415. How do you refresh test environment DBs from prod in Azure SQL?
416. Azure SQL maintenance caused downtime. How do you minimize?
417. How do you configure read scale-out replicas?
418. You must enforce row-level security in Azure SQL. How?
419. How do you detect blocking sessions in Azure SQL?
420. Azure SQL pricing increased. How do you optimize costs?
421. How do you configure log shipping equivalent in Azure SQL?
422. Azure SQL shows query timeout errors. How do you resolve?
423. How do you configure failover group automatic testing?
424. Azure SQL restore exceeds allocated space. What next?
425. You must implement GDPR compliance in Azure SQL. How?
426. Azure SQL job scheduling is missing. What's your alternative?
427. How do you migrate SSIS to Azure SQL MI?
428. How do you configure Azure SQL alerts in Log Analytics?
429. Azure SQL firewall rules reset. Why?
430. How do you configure auditing to Azure Storage?
431. Azure SQL query optimizer regression occurs. How do you fix?
432. How do you configure connection resilience in Azure SQL?
433. A developer needs database mail in Azure SQL. What's the alternative?
434. Azure SQL maintenance window overlaps with workload. How do you fix?
435. How do you configure failover for Azure Managed Instance to on-prem?
436. A query in Azure SQL consumes too many worker threads. How do you tune?
437. How do you manage index fragmentation in Azure SQL?

438. Azure SQL elastic pool performance degrades. What do you do?
439. How do you configure cross-region geo-restore?
440. Query store plan forcing fails. Why?
441. How do you configure alerting for DTU saturation?
442. Azure SQL audit shows failed logins. What do you do?
443. A database restore needs to rename DB. How do you handle in Azure SQL?
444. How do you manage retention for long-term backups in Azure SQL?
445. Azure SQL connections drop during failover. How do you minimize?
446. How do you handle schema migration automation in Azure SQL?
447. A large query causes tempdb contention. How do you fix in Azure SQL?
448. How do you implement hybrid replication to Azure SQL MI?
449. Azure SQL shows excessive storage usage. How do you analyze?
450. Azure SQL connection auditing to Event Hub fails. How do you fix?
451. How do you optimize bulk load performance in Azure SQL?
452. Azure SQL login sync fails with Azure AD group. Why?
453. How do you implement service endpoints for Azure SQL?
454. Azure SQL DB reached max size. How do you fix?
455. How do you integrate Azure SQL with Managed Identity?
456. A query triggers excessive IO in Azure SQL. How do you tune?
457. Azure SQL DTU model migration to vCore fails. Why?
458. How do you configure geo-failover DR drills in Azure SQL?
459. Azure SQL update breaks compatibility level. How do you fix?
460. How do you refresh Azure SQL from bacpac exports?
461. Azure SQL elastic query fails. How do you debug?
462. How do you configure partitioning in Azure SQL?
463. Azure SQL performance drops during scaling. Why?
464. How do you configure failover group listener in Azure SQL?
465. A backup restore in Azure SQL is incomplete. How do you check?
466. How do you enforce auditing for all Azure SQL DBs centrally?
467. Azure SQL query store capture fails. What's your fix?
468. How do you enable data masking in Azure SQL?
469. Azure SQL firewall blocks Power BI access. How do you fix?
470. How do you secure secrets for Azure SQL connections?
471. Azure SQL elastic pool hits connection limits. How do you fix?
472. How do you configure Transparent Data Encryption in Azure SQL?
473. Azure SQL replication to on-prem fails. What's your fix?
474. How do you monitor deadlocks in Azure SQL?
475. Azure SQL performance drops during maintenance. How do you mitigate?
476. How do you restore Azure Managed Instance to a new VNet?
477. Azure SQL long-running queries fail at timeout. How do you extend?
478. How do you set up continuous auditing in Azure SQL?
479. Azure SQL firewall allows unwanted IPs. How do you fix?
480. How do you tune queries using Query Performance Insights?
481. Azure SQL DB dropped accidentally. How do you recover?
482. How do you configure zone redundancy for Azure SQL?
483. Azure SQL DB high log write waits. How do you tune?
484. How do you implement CDC in Azure SQL MI?
485. Azure SQL backup to Blob Storage fails. How do you debug?
486. How do you manage permissions across 100 Azure SQL DBs?
487. Azure SQL MI AG failover fails. What do you check?

- 488. How do you monitor query wait stats in Azure SQL?
- 489. Azure SQL reports blocking by system processes. Why?

Section C: AWS EC2 with SQL Server – Continued (Q71–Q110)

- 71. How do you automate SQL Server patching on EC2 using SSM (Systems Manager)?
- 72. SQL Server EC2 instance keeps rebooting after patching. How do you roll back?
- 73. How do you replicate data from EC2 SQL Server to RDS?
- 74. How do you configure AlwaysOn cluster nodes across 3 AZs in EC2?
- 75. An EC2 SQL Server IOPS limit is throttling performance. How do you resolve?
- 76. How do you test a SQL Server DR failover from EC2 in Region A to Region B?
- 77. Your SQL backups from EC2 to S3 are failing due to IAM issues. How do you fix?
- 78. EC2 SQL instance is under DDoS attack. How do you mitigate?
- 79. How do you integrate EC2 SQL Server backups with AWS Backup service?
- 80. How do you set up CloudWatch Logs to capture SQL Error Logs automatically?
- 81. How do you design EC2 SQL HA with shared storage (FSx or EBS Multi-Attach)?
- 82. Your SQL Server running on EC2 is migrating to Graviton-based instance. Any risks?
- 83. SQL Server AlwaysOn EC2 cluster shows “quorum lost.” How do you recover?
- 84. How do you implement transparent encryption for SQL backups stored in S3?
- 85. A SQL Server EC2 migration is stuck due to network bottleneck. How do you speed up?
- 86. How do you configure hybrid replication (on-prem → EC2 SQL → RDS)?
- 87. SQL Server EC2 security group is too permissive. How do you harden access?
- 88. How do you configure KMS-encrypted EBS volumes for SQL data/log files?
- 89. Your SQL EC2 is part of an Auto Scaling Group by mistake. What’s the impact?
- 90. How do you migrate EC2 SQL Server to RDS with minimal downtime?
- 91. An EC2 instance backup restore corrupted the SQL data file. How do you recover?
- 92. Your EC2 SQL instance is running on HDD. How do you migrate to SSD with zero downtime?
- 93. How do you set up Performance Insights equivalent for SQL Server on EC2?
- 94. How do you secure SQL Server ports (1433) in EC2 without breaking apps?
- 95. How do you configure AWS Secrets Manager for SQL logins in EC2?
- 96. SQL Server licensing cost on EC2 is too high. What optimization options do you have?
- 97. A SQL Server EC2 Windows update breaks cluster services. What’s your rollback?
- 98. How do you ensure EC2 SQL backups are compliant with HIPAA?
- 99. How do you automate SQL Server failover between EC2 and on-prem DR site?
- 100. A SQL Server EC2 instance has corrupted boot volume. How do you restore?
- 101. How do you use AWS DMS to migrate EC2 SQL data to Aurora?
- 102. SQL Server EC2 tempdb keeps filling EBS disk. How do you tune?
- 103. How do you configure DNS failover for SQL AlwaysOn cluster in EC2?
- 104. Your SQL EC2 instance is hitting max concurrent connections. What’s your action?
- 105. How do you patch EC2 SQL cluster nodes across multiple regions?
- 106. EC2 SQL backups are taking too long. How do you optimize backup strategy?
- 107. How do you configure CloudFormation template for SQL Server EC2 deployment?
- 108. SQL Server EC2 RPO/RTO needs are not met. How do you redesign DR?
- 109. How do you migrate SQL Server EC2 instance from Windows 2012 to Windows 2022?
- 110. How do you configure hybrid AlwaysOn between EC2 and Azure SQL MI?

Section D: Azure SQL Database & Managed Instance – 120 Questions

General Scenarios (Q111–Q140)

111. Your Azure SQL Database is throttling queries due to DTU cap. How do you resolve?
112. How do you implement cross-database queries in Azure SQL Database?
113. Your Azure SQL DB is in Paused state (serverless). How do you force resume?
114. How do you configure active geo-replication for DR?
115. A user dropped a table. How do you recover it?
116. How do you migrate on-prem SQL Server to Azure SQL MI with minimal downtime?
117. How do you troubleshoot slow query performance in Azure SQL?
118. How do you handle row-level security migration from on-prem SQL to Azure SQL?
119. Your Azure SQL DB hit max size (4TB). What's your plan?
120. How do you configure VNet integration for Azure SQL MI?
121. Your backup retention requirement is 10 years. How do you configure in Azure SQL?
122. How do you troubleshoot frequent throttling errors (Error 10928/10929)?
123. How do you configure AAD authentication for Azure SQL DB?
124. Your Azure SQL DB is accidentally deleted. How do you restore?
125. How do you monitor Azure SQL with Log Analytics?
126. How do you configure TDE with custom keys in Key Vault?
127. How do you migrate SQL Server Agent jobs to Azure SQL?
128. How do you configure failover groups across regions for MI?
129. How do you resolve "long running transaction" in Azure SQL DB?
130. Your query plan cache is bloated. How do you clear it?
131. How do you troubleshoot latency between app and Azure SQL MI?
132. Your data sync between Azure SQL DBs is failing. How do you fix?
133. How do you configure auditing logs to Storage Account?
134. A query times out only in Azure SQL but works on-prem. How do you fix?
135. How do you configure zone redundancy for Azure SQL DB?
136. How do you reduce costs for an underutilized Azure SQL MI?
137. How do you enable automatic tuning features in Azure SQL?
138. How do you configure Always Encrypted columns in Azure SQL?
139. A developer needs CLR functions in Azure SQL DB. How do you handle?
140. How do you migrate SQL Server logins to Azure SQL?

HA/DR Scenarios (Q141–Q170)

141. How do you test geo-failover for Azure SQL DB?
142. How do you recover if both primary and replica Azure SQL DBs fail?
143. How do you achieve RPO < 5 min in Azure SQL MI?
144. Your geo-secondary is lagging behind by 1 hour. How do you troubleshoot?
145. How do you design DR for 50+ Azure SQL DBs?
146. How do you configure Auto-Failover Groups for Managed Instance?
147. What's your plan if an entire Azure region goes down?
148. How do you validate PITR restores regularly?
149. How do you configure failover between East US and West US?
150. How do you automate DR drills for Azure SQL?
151. How do you design HA for Azure SQL DB Business Critical tier?
152. How do you perform failback after geo-failover?
153. What steps do you take if geo-restore is not working?
154. How do you ensure SLA compliance for Azure SQL HA?
155. Your secondary is in read-only mode. How do you offload reporting?

156. How do you test failover groups without affecting production?
157. How do you configure auto-failover with app connection string changes?
158. How do you mitigate Azure SQL planned maintenance downtime?
159. How do you reduce RTO from 1 hour to 5 minutes?
160. What's your plan if Point-in-Time Restore doesn't have required LSNs?
161. How do you configure backup to blob storage for compliance?
162. Your DR test shows logins missing after failover. How do you fix?
163. How do you design multi-region failover for MI?
164. How do you handle failover testing for an elastic pool?
165. How do you restore encrypted DB backups in Azure SQL MI?
166. How do you configure Backup Long-Term Retention (LTR)?
167. Your PITR restore is failing due to "storage full." How do you fix?
168. How do you validate backups in Azure SQL DB automatically?
169. How do you handle GDPR compliance in Azure SQL backups?
170. How do you automate HA/DR health checks with PowerShell/CLI?

Security & Performance (Q171–Q200)

171. How do you configure firewall rules for Azure SQL securely?
172. How do you configure AAD-only authentication?
173. How do you prevent SQL injection in Azure SQL apps?
174. How do you configure private endpoints for Azure SQL DB?
175. How do you configure row-level security in Azure SQL DB?
176. Your DTU utilization is at 95%. How do you tune workloads?
177. How do you migrate SQL jobs using Elastic Jobs in Azure?
178. How do you configure SQL Auditing to Event Hubs?
179. Your query is running slow due to missing statistics. How do you fix?
180. How do you configure workload management in MI?
181. How do you enforce TLS 1.2 connections only?
182. How do you troubleshoot excessive deadlocks in Azure SQL?
183. How do you configure Query Store for tuning?
184. How do you track queries causing resource governance throttling?
185. How do you configure custom alerts for blocking sessions?
186. How do you implement Always Encrypted with secure enclaves?
187. How do you configure Azure Monitor metrics for SQL?
188. How do you optimize costs with elastic pools?
189. Your app requires CLR, but Azure SQL DB doesn't support. Workaround?
190. How do you configure customer-managed TDE keys?
191. How do you enforce data masking for developers?
192. How do you troubleshoot "Database copy failed" error?
193. How do you optimize queries hitting tempdb in MI?
194. How do you handle query regression after automatic tuning?
195. How do you configure auditing for HIPAA in Azure SQL?
196. How do you enable ledger in Azure SQL DB for blockchain-style auditing?
197. How do you configure Azure Policy to enforce SQL security?
198. Your MI is using excessive tempdb. How do you mitigate?
199. How do you configure cross-region read scale-out replicas?
200. How do you validate network latency between app and Azure SQL?

1. SQL Server Installation & Configuration (40)

1. How do you decide between a default instance and a named instance during installation?
2. What are the key SQL Server services installed and their purpose?
3. How do you configure SQL Server memory after installation?
4. What is the importance of TempDB configuration during installation?
5. How do you configure SQL Server to use instant file initialization?
6. How do you decide on collation settings during installation?
7. What are the steps to configure SQL Server to use mixed authentication mode?
8. How do you configure default file locations for data and log files?
9. What is the difference between SQL Server Developer Edition and Enterprise Edition?
10. How do you configure SQL Server network protocols (TCP/IP, Named Pipes)?
11. How do you install SQL Server on a cluster?
12. What's the difference between stand-alone and clustered SQL Server installation?
13. How do you apply a cumulative update or service pack in SQL Server?
14. How do you check SQL Server version and patch level?
15. What is the role of SQL Server Configuration Manager?
16. How do you configure SQL Server startup parameters?
17. How do you enable or disable SQL Server Agent?
18. What are the security considerations during installation?
19. How do you install SQL Server using command line / unattended installation?
20. How do you configure SQL Server to use SSL encryption?
21. What are the system databases and their purposes?
22. How do you configure multiple instances on the same server?
23. What is the significance of SQL Server default ports?
24. How do you configure SQL Server to listen on a different port?
25. What are SQL Server surface area configurations?
26. How do you enable/disable SQL Server Browser service?
27. How do you check which features are installed in SQL Server?
28. What are startup accounts and their security implications?
29. How do you migrate SQL Server installation to a new server?
30. How do you configure SQL Server for best practices immediately after installation?
31. How do you verify SQL Server setup logs for installation issues?
32. How do you handle failed SQL Server installation rollback?
33. How do you plan SQL Server capacity before installation?
34. What is the difference between SQL Server Express and Standard editions?
35. How do you change SQL Server service accounts post-installation?
36. How do you configure SQL Server with Active Directory authentication?
37. How do you install SQL Server command line utilities?
38. How do you configure client aliases in SQL Server?
39. How do you configure SQL Server for Kerberos authentication?
40. What are best practices for installation in production?

2. Database Creation & Management (30)

41. How do you create a new database with specific filegroups?
42. What are primary, secondary, and log files in a database?
43. How do you set database compatibility level?
44. How do you configure auto-growth settings for data and log files?
45. What is the difference between SIMPLE, FULL, and BULK_LOGGED recovery models?
46. How do you shrink data/log files and why should you avoid it?

47. How do you move database files to a different location?
48. What is the importance of filegroups in large databases?
49. How do you check database properties using T-SQL?
50. What is a contained database?
51. How do you configure database options like AUTO_CLOSE, AUTO_SHRINK?
52. How do you set up database ownership?
53. How do you rename a database safely?
54. What are database snapshots and when would you use them?
55. How do you handle database corruption in system databases?
56. How do you attach/detach a database?
57. What are orphaned users and how do you fix them?
58. How do you move a database from one server to another?
59. How do you enable TDE (Transparent Data Encryption)?
60. How do you check database size and growth trends?
61. How do you configure multiple data files for TempDB?
62. What's the difference between database restore and attach?
63. How do you handle database compatibility for upgraded versions?
64. What is DBCC CHECKDB and when should you run it?
65. How do you enforce database collation settings?
66. How do you configure database-level auditing?
67. How do you manage database-level encryption keys?
68. What are partial contained databases?
69. How do you clone a database without downtime?
70. How do you check active connections to a database?

3. Backup & Recovery (40)

71. What are the different types of backups in SQL Server?
72. What is the difference between differential and log backups?
73. How do you take a tail-log backup?
74. How do you verify backups are valid?
75. What is backup compression and when to use it?
76. How do you automate backup strategies with SQL Agent?
77. What is COPY_ONLY backup?
78. How do you restore a database to a different point in time?
79. What is a piecemeal restore?
80. How do you restore a database with NORECOVERY vs STANDBY?
81. What is striped backup?
82. How do you perform page-level restore?
83. How do you restore system databases (master, msdb)?
84. How do you handle backup failures?
85. What is backup checksum?
86. How do you test database restore on a DR site?
87. What is log chain and how do you maintain it?
88. What are file/filegroup backups?
89. How do you move backups to cloud (Azure, AWS)?
90. What are SQL Server backup best practices?
91. How do you schedule log backups without breaking the log chain?
92. What is the difference between RESTORE WITH RECOVERY and RESTORE WITH NORECOVERY?
93. How do you recover accidentally deleted data?
94. What is point-in-time recovery?

95. How do you handle backup retention policies?
96. What is VDI backup and how does it work?
97. How do you secure backup files?
98. What is differential base in backups?
99. How do you restore encrypted backups?
100. How do you check backup metadata in msdb?
101. What is backup expiration?
102. How do you manage backup file cleanup?
103. How do you configure backup priority in AlwaysOn AG?
104. How do you take backups in log-shipping setup?
105. How do you troubleshoot slow backups?
106. How do you encrypt backups with certificates?
107. How do you restore from striped backup files?
108. What is block-level backup vs native SQL backup?
109. How do you take backup of large databases efficiently?
110. How do you verify log sequence number (LSN) during restore?

4. High Availability & Disaster Recovery (HADR) (60)

111. What are the different HA/DR options in SQL Server?
112. What is the difference between AlwaysOn Availability Groups and Failover Cluster Instances?
113. How do you configure log shipping?
114. What are the prerequisites for setting up database mirroring?
115. How do you configure synchronous vs asynchronous AG?
116. What are automatic failover conditions in Availability Groups?
117. How do you check synchronization status of replicas?
118. How do you troubleshoot log shipping latency?
119. What is quorum in Windows clustering?
120. What is a listener in Availability Groups?
121. How do you add/remove databases from Availability Group?
122. What is the difference between readable secondary and non-readable secondary replicas?
123. How do you configure read-only routing in AG?
124. How do you monitor failover events in AlwaysOn?
125. How do you configure witness server in database mirroring?
126. What are different failover modes in database mirroring?
127. What is the role of MSDTC in SQL clustering?
128. How do you patch SQL Server in an AG setup with minimal downtime?
129. How do you test DR failover for AG without downtime?
130. What are split-brain scenarios in clustering?
131. What is seeding mode in AG?
132. How do you configure AG automatic seeding?
133. What are potential causes of AG synchronization issues?
134. How do you troubleshoot AG failover delays?
135. What is endpoint in database mirroring/AG?
136. How do you secure endpoints for mirroring/AG?
137. What are common issues with log shipping and how to fix them?
138. How do you perform role switch in database mirroring?
139. How do you configure distributed AGs?
140. What is the difference between FCI and AG in licensing perspective?
141. How do you monitor log shipping using msdb tables?
142. What are differences between standard and enterprise edition HA options?

143. How do you implement failover cluster across data centers?
144. What are quorum modes (Node Majority, Node+File Share, Node+Disk)?
145. How do you validate AG setup health using T-SQL?
146. How do you remove mirroring from a database?
147. How do you reinitialize log shipping secondary?
148. What are the steps to restore AG database to a standalone server?
149. How do you add a replica in distributed AG?
150. How do you test failover without impacting production?
151. How do you configure AG on Linux SQL Server?
152. How do you monitor AG latency in real time?
153. What are differences between synchronous commit and asynchronous commit?
154. How do you plan RTO and RPO in SQL Server HA?
155. How do you handle TempDB in AlwaysOn AG?
156. What is the role of DNS in failover cluster?
157. How do you configure SQL Server with Azure Site Recovery?
158. What is stretched cluster and when to use it?
159. How do you configure automatic page repair in AG?
160. How do you backup databases in AG setup?
161. How do you test disaster recovery drill?
162. What are AG dashboard warning signs?
163. How do you force failover in database mirroring?
164. What happens if quorum is lost in cluster?
165. How do you add a new database to log shipping setup?
166. What is cross-database transaction issue in AG?
167. How do you monitor failover cluster logs?
168. How do you set preferred replica for backups?
169. What is the impact of network latency on synchronous AG?
170. How do you configure distributed transaction support in AG?

5. Performance Tuning & Optimization (80)

171. How do you identify long-running queries in SQL Server?
172. What is execution plan and how do you analyze it?
173. How do you capture deadlocks?
174. What are missing index recommendations in SQL Server?
175. How do you check fragmentation levels?
176. What is the difference between clustered and non-clustered indexes?
177. How do you implement index maintenance strategy?
178. How do you troubleshoot blocking sessions?
179. How do you detect tempdb contention issues?
180. What is parameter sniffing and how to fix it?
181. How do you use Query Store for performance troubleshooting?
182. How do you capture query performance using Extended Events?
183. How do you calculate query cost?
184. What is a covering index?
185. How do you decide fill factor for indexes?
186. What is the impact of implicit conversions on performance?
187. How do you monitor CPU bottlenecks in SQL Server?
188. How do you troubleshoot memory pressure issues?
189. What are wait statistics and how do you analyze them?
190. How do you capture IO bottlenecks?

191. How do you identify top resource consuming queries?
192. What is plan cache and how to clear it?
193. How do you identify ad-hoc queries causing performance issues?
194. How do you troubleshoot parameter sniffing issues?
195. How do you handle statistics updates?
196. How do you check for outdated statistics?
197. What is filtered index and when to use it?
198. How do you use hints for query optimization?
199. How do you configure MAXDOP for optimal performance?
200. What is the role of Resource Governor in performance management?
201. How do you check buffer cache hit ratio?
202. How do you capture slow queries using Profiler/Extended Events?
203. How do you monitor latch contention?
204. How do you troubleshoot excessive recompilations?
205. What is plan regression and how to fix it?
206. How do you force a query to use a specific execution plan?
207. What are index usage stats and how to use them?
208. How do you troubleshoot high disk latency?
209. How do you analyze tempdb usage?
210. How do you fix CXPACKET waits?
211. How do you decide between heap and clustered index?
212. How do you capture queries using DMVs?
213. What is the difference between actual and estimated execution plan?
214. How do you reduce recompilation overhead?
215. What is batch mode processing?
216. How do you troubleshoot excessive memory grants?
217. How do you capture top waits in SQL Server?
218. What are SOS_SCHEDULER_YIELD waits?
219. How do you troubleshoot high PAGEIOLATCH_SH waits?
220. How do you analyze spinlock contention?
221. How do you optimize queries with OR conditions?
222. How do you handle queries using functions in WHERE clause?
223. What is statistics histogram and how to interpret it?
224. How do you fix queries not using indexes?
225. What is temp table vs table variable performance difference?
226. How do you troubleshoot excessive plan cache bloat?
227. How do you manage parameterized queries for performance?
228. What is adaptive query processing?
229. How do you tune queries with dynamic SQL?
230. What is query hint OPTIMIZE FOR UNKNOWN?
231. How do you monitor query performance over time?
232. How do you implement index partitioning for performance?
233. How do you analyze memory clerks?
234. What is stale plan and how to fix it?
235. How do you analyze query compile time?
236. How do you troubleshoot excessive blocking caused by statistics update?
237. How do you use sp_whoisactive for troubleshooting?
238. How do you capture queries causing tempdb spills?
239. What is memory grant feedback?
240. How do you troubleshoot queries with scalar UDFs?

- 241. What is delayed durability and how does it impact performance?
- 242. How do you use Query Store forced plans?
- 243. How do you troubleshoot excessive ghost records?
- 244. What is parameter embedding optimization?
- 245. How do you optimize large insert operations?
- 246. How do you monitor plan cache reuse?
- 247. How do you troubleshoot excessive CPU by query compilation?
- 248. How do you monitor batch requests/sec?
- 249. How do you detect plan cache eviction?
- 250. How do you tune queries with wide tables?

6. Security & Compliance (40)

- 251. How do you create SQL Server logins and users?
- 252. What is the difference between login and user?
- 253. How do you enforce password policies in SQL Server?
- 254. How do you configure server roles?
- 255. How do you configure database roles?
- 256. How do you grant/revoke permissions?
- 257. What is the principle of least privilege in SQL Server?
- 258. How do you monitor failed login attempts?
- 259. How do you configure SQL Server Audit?
- 260. How do you configure C2 auditing?
- 261. How do you configure login triggers?
- 262. How do you detect SQL injection attempts?
- 263. What is contained database authentication?
- 264. How do you configure row-level security?
- 265. How do you implement column-level encryption?
- 266. What is Dynamic Data Masking?
- 267. How do you configure TDE?
- 268. How do you backup and restore certificates?
- 269. How do you manage service master key?
- 270. How do you rotate encryption keys?
- 271. How do you enforce SSL encryption for connections?
- 272. What is Kerberos authentication in SQL Server?
- 273. How do you implement Always Encrypted?
- 274. What are security implications of xp_cmdshell?
- 275. How do you configure SQL injection protection at DB level?
- 276. How do you detect orphaned users?
- 277. How do you enforce data access compliance for GDPR?
- 278. How do you configure login auditing?
- 279. How do you manage guest user account?
- 280. How do you secure linked servers?
- 281. How do you configure application roles?
- 282. How do you implement proxy accounts in SQL Agent?
- 283. How do you configure permissions for cross-database queries?
- 284. How do you monitor privileged account usage?
- 285. What is separation of duties in SQL Server?
- 286. How do you configure SQL Server Firewall rules?
- 287. How do you enforce network encryption using certificates?
- 288. How do you monitor schema changes?

- 289. What are security best practices for SQL Server?
- 290. How do you perform security penetration testing on SQL Server?

7. Monitoring & Maintenance (50)

- 291. How do you configure maintenance plans?
- 292. How do you schedule index rebuild/reorganize?
- 293. How do you monitor SQL Agent jobs?
- 294. How do you troubleshoot failed jobs?
- 295. How do you capture error logs?
- 296. What are default SQL Server trace files?
- 297. How do you purge error logs?
- 298. How do you configure custom alerts in SQL Server Agent?
- 299. How do you monitor SQL Server performance using PerfMon?
- 300. How do you configure Data Collector?
- 301. How do you monitor SQL Server using DMVs?
- 302. What is Policy-Based Management?
- 303. How do you configure Database Mail?
- 304. How do you send alerts using DB mail?
- 305. How do you monitor blocking using alerts?
- 306. How do you configure operator notifications?
- 307. How do you monitor SQL Server uptime?
- 308. How do you configure health check jobs?
- 309. How do you check SQL Server startup time?
- 310. How do you configure Central Management Server?
- 311. How do you use Multi-Server jobs in SQL Server Agent?
- 312. How do you configure alerts for severity-level errors?
- 313. How do you monitor SQL Server memory usage trends?
- 314. How do you capture IO statistics for each database file?
- 315. How do you configure automatic index maintenance scripts?
- 316. How do you configure custom health checks for SQL Server?
- 317. How do you monitor SQL Server job history for long-running jobs?
- 318. How do you archive historical job execution data?
- 319. How do you configure and monitor SQL Server Extended Events?
- 320. How do you monitor SQL Server using third-party tools like SentryOne or RedGate?
- 321. How do you monitor SQL Server AlwaysOn AG health?
- 322. How do you monitor SQL Server Failover Cluster status?
- 323. How do you configure alerts for failed logins?
- 324. How do you monitor index fragmentation regularly?
- 325. How do you automate DBCC CHECKDB execution?
- 326. How do you monitor log file size growth?
- 327. How do you configure alerts for low disk space?
- 328. How do you monitor SQL Server deadlocks automatically?
- 329. How do you capture SQL Server deadlock graphs?
- 330. How do you configure server-side traces for troubleshooting?
- 331. How do you monitor SQL Server replication health?
- 332. How do you check the last successful DBCC CHECKDB run?
- 333. How do you configure alerts for high CPU usage?
- 334. How do you track schema changes in SQL Server?
- 335. How do you monitor TempDB usage growth?
- 336. How do you configure index usage reports?

- 337. How do you monitor SQL Server security audits?
- 338. How do you configure alerts for blocked processes?
- 339. How do you use Resource Governor to monitor workload distribution?
- 340. How do you configure alerts for failed backups?

8. Troubleshooting & Real-Time Scenarios (70)

- 341. How do you troubleshoot SQL Server that won't start?
- 342. How do you recover from a corrupted master database?
- 343. How do you troubleshoot login failures?
- 344. How do you fix orphaned users after database restore?
- 345. How do you recover from "suspect" database status?
- 346. How do you troubleshoot SQL Server high CPU usage?
- 347. How do you troubleshoot SQL Server memory pressure?
- 348. How do you troubleshoot disk IO bottlenecks in SQL Server?
- 349. How do you troubleshoot TempDB full errors?
- 350. How do you troubleshoot SQL Server blocking chains?
- 351. How do you capture and resolve deadlocks?
- 352. How do you troubleshoot long-running jobs in SQL Agent?
- 353. How do you recover deleted records from transaction logs?
- 354. How do you troubleshoot high VLF counts in transaction log?
- 355. How do you troubleshoot excessive recompilation issues?
- 356. How do you troubleshoot SQL Server service not starting due to port conflicts?
- 357. How do you fix "Login failed for user" error?
- 358. How do you troubleshoot failed backups?
- 359. How do you recover missing or deleted database files?
- 360. How do you troubleshoot database stuck in "restoring" state?
- 361. How do you troubleshoot AG synchronization issues?
- 362. How do you fix log shipping secondary behind primary?
- 363. How do you troubleshoot mirroring connection errors?
- 364. How do you troubleshoot failover cluster resource failures?
- 365. How do you handle SQL Server stuck in recovery during startup?
- 366. How do you troubleshoot replication latency issues?
- 367. How do you fix distribution database growth in replication?
- 368. How do you troubleshoot snapshot replication failures?
- 369. How do you troubleshoot merge replication conflicts?
- 370. How do you recover from dropped indexes in production?
- 371. How do you fix queries running slow after an upgrade?
- 372. How do you troubleshoot excessive TempDB spills?
- 373. How do you troubleshoot excessive waits in SQL Server?
- 374. How do you recover SQL Server login passwords?
- 375. How do you troubleshoot database encryption issues?
- 376. How do you fix SQL Agent not starting?
- 377. How do you troubleshoot jobs stuck in "Executing" status?
- 378. How do you fix failed SQL Server patches?
- 379. How do you recover master key if password is lost?
- 380. How do you fix SQL Server reporting "Could not allocate space for object in tempdb"?
- 381. How do you troubleshoot failed cluster failover?
- 382. How do you troubleshoot AG listener connection issues?
- 383. How do you troubleshoot slow log restores in DR testing?
- 384. How do you fix SQL Server database stuck in "single-user" mode?

- 385. How do you troubleshoot login timeout issues?
- 386. How do you fix blocking caused by ghost cleanup?
- 387. How do you troubleshoot excessive tempdb version store usage?
- 388. How do you troubleshoot SQL Server "access denied" on backup file path?
- 389. How do you fix replication reinitialization issues?
- 390. How do you troubleshoot failed linked server connections?
- 391. How do you handle SQL Server "transaction log full" error?
- 392. How do you recover from "system tables are corrupted" error?
- 393. How do you troubleshoot "SQL Server taking too long to stop"?
- 394. How do you handle SQL Server running out of worker threads?
- 395. How do you troubleshoot "could not create tempdb" error?
- 396. How do you troubleshoot issues with xp_cmdshell execution?
- 397. How do you recover accidentally dropped database in production?
- 398. How do you troubleshoot AlwaysOn AG failover loops?
- 399. How do you troubleshoot cluster witness failures?
- 400. How do you recover data from damaged MDF file?
- 401. How do you troubleshoot blocked backups due to active transactions?
- 402. How do you troubleshoot failed SQL login from application servers?
- 403. How do you troubleshoot login storms affecting performance?
- 404. How do you troubleshoot queries running differently on primary and secondary replicas?
- 405. How do you troubleshoot high SOS_SCHEDULER_YIELD waits?
- 406. How do you troubleshoot SQL Server "insufficient system memory" error?
- 407. How do you troubleshoot sudden growth of transaction log file?
- 408. How do you troubleshoot "TempDB log full" error?
- 409. How do you troubleshoot replication agents not starting?
- 410. How do you troubleshoot excessive checkpoint activity?

9. Replication (20)

- 411. What are the types of replication in SQL Server?
- 412. What is snapshot replication and when to use it?
- 413. What is transactional replication and when to use it?
- 414. What is merge replication and when to use it?
- 415. How do you configure replication agents?
- 416. What are replication latency monitoring techniques?
- 417. How do you reinitialize a replication subscription?
- 418. How do you monitor replication conflicts?
- 419. How do you add a new article to replication?
- 420. How do you handle identity columns in replication?
- 421. How do you configure replication over WAN?
- 422. How do you troubleshoot replication agent failures?
- 423. How do you check replication metadata tables?
- 424. How do you clean up distribution database growth?
- 425. How do you monitor replication using Replication Monitor?
- 426. What are limitations of replication in SQL Server?
- 427. How do you configure peer-to-peer replication?
- 428. How do you monitor replication topology?
- 429. How do you upgrade replication to a new SQL version?
- 430. How do you configure replication security?

10. SQL Server Patching, Migration & Upgrades (30)

431. How do you plan SQL Server version upgrades?
432. How do you check application compatibility before upgrade?
433. How do you perform in-place vs side-by-side upgrade?
434. How do you migrate SQL Server databases to new hardware?
435. How do you upgrade AlwaysOn AG to new SQL version?
436. How do you migrate SQL logins between servers?
437. How do you migrate SQL Server jobs to new instance?
438. How do you migrate linked servers?
439. How do you migrate replication setup?
440. How do you migrate SSIS packages to new SQL version?
441. How do you handle deprecated features during upgrade?
442. How do you check SQL Server upgrade readiness reports?
443. How do you migrate certificates and keys to new server?
444. How do you upgrade SQL Server running in a failover cluster?
445. How do you upgrade SQL Server running on AG?
446. How do you upgrade SQL Server running replication?
447. How do you test application performance post-upgrade?
448. How do you rollback failed SQL Server upgrade?
449. How do you migrate SQL Server databases to cloud (Azure, AWS)?
450. How do you migrate SQL Server databases with minimal downtime?
451. How do you configure log shipping for migration?
452. How do you configure backup/restore migration strategy?
453. How do you use Distributed AG for migration?
454. How do you validate migration success?
455. How do you upgrade SQL Server with minimal downtime?
456. How do you migrate jobs/operators/proxies?
457. How do you migrate SQL Server instance-level configurations?
458. How do you migrate SQL Server linked reports?
459. How do you perform SQL Server edition downgrade?
460. How do you migrate SQL Server collation settings?

11. Cloud, DevOps & Automation (20)

461. How do you deploy SQL Server in Azure VM?
462. What is the difference between Azure SQL Database and SQL Server on VM?
463. How do you configure SQL Server backups to Azure Blob Storage?
464. How do you configure automated patching in Azure SQL Managed Instance?
465. How do you configure automated backups in AWS RDS SQL Server?
466. How do you automate SQL Server deployment with PowerShell?
467. How do you monitor SQL Server using Azure Monitor?
468. How do you configure alerts in AWS CloudWatch for SQL Server?
469. How do you use ARM templates to deploy SQL Server?
470. How do you configure CI/CD for SQL Server deployments?
471. How do you use dbatools PowerShell module for SQL Server automation?
472. How do you configure auto-scaling for SQL Server workloads?
473. How do you migrate SQL Server to Azure SQL Managed Instance?
474. How do you configure Azure Site Recovery for SQL Server?
475. How do you configure SQL Server in Kubernetes containers?
476. How do you configure hybrid disaster recovery with SQL Server?
477. How do you configure Azure SQL failover groups?

- 478. How do you configure automated index maintenance with PowerShell?
- 479. How do you configure SQL Server monitoring with Grafana/Prometheus?
- 480. How do you implement Infrastructure as Code for SQL Server deployments?

How do you troubleshoot connection pooling issues in Azure SQL DB?

- 431. How do you troubleshoot slow queries when Query Store is disabled?
- 432. How do you troubleshoot failed TDE key rotations in SQL DB?
- 433. How do you troubleshoot login failures due to Azure AD misconfiguration?
- 434. How do you troubleshoot replication lag in geo-replicated databases?
- 435. How do you troubleshoot data sync errors between Azure SQL DB and on-prem?
- 436. How do you troubleshoot memory pressure in Business Critical tier?
- 437. How do you troubleshoot queries spilling to tempdb?
- 438. How do you troubleshoot distributed query failures with external tables?
- 439. How do you troubleshoot blocked system views in Azure SQL DB?
- 440. How do you troubleshoot firewall rule conflicts across subscriptions?
- 441. How do you troubleshoot queries failing due to MAXDOP limitations?
- 442. How do you troubleshoot row version store space exhaustion?
- 443. How do you troubleshoot high waits on SOS_SCHEDULER_YIELD in SQL DB?
- 444. How do you troubleshoot missing execution plans in Query Store?
- 445. How do you troubleshoot managed identity authentication failures?
- 446. How do you troubleshoot TLS handshake timeouts from client apps?
- 447. How do you troubleshoot database stuck in "Restoring" state?
- 448. How do you troubleshoot elastic pool DTU starvation?
- 449. How do you troubleshoot deadlocks between multiple tenants?

Disaster Recovery & Business Continuity (Advanced)

- 450. How do you test failover groups during Azure regional outage simulation?
- 451. How do you configure DR drills without impacting production?
- 452. How do you validate RTO/RPO in geo-replication scenarios?
- 453. How do you recover from failover group misconfiguration?
- 454. How do you automate DR validation across multiple databases?
- 455. How do you configure alerts for replication lag in DR setups?
- 456. How do you handle schema drift between primary and geo-secondary?
- 457. How do you restore databases to isolated subscription during DR?
- 458. How do you simulate accidental region-wide deletion recovery?
- 459. How do you configure DNS cutover validation during failover?

Automation & DevOps

- 460. How do you integrate Azure SQL DB deployments in CI/CD pipelines?
- 461. How do you automate schema drift detection with pipelines?
- 462. How do you automate scaling rules using ARM templates?
- 463. How do you use Terraform to deploy Azure SQL DB at scale?
- 464. How do you configure automated security baseline checks in DevOps?
- 465. How do you run automated performance tests after schema deployment?
- 466. How do you manage connection strings securely in DevOps pipelines?
- 467. How do you configure Blue/Green deployments for schema changes?
- 468. How do you implement automated rollback for failed deployments?
- 469. How do you automate data masking in pipeline refreshes?
- 470. How do you enforce naming conventions in Azure SQL DB deployments?

Monitoring & Observability (Advanced)

- 471. How do you configure proactive anomaly detection using Azure Monitor?
- 472. How do you integrate SQL Insights metrics with Grafana?
- 473. How do you monitor query performance across elastic pools centrally?
- 474. How do you configure alerts on unusual login patterns?
- 475. How do you monitor top resource-consuming queries at scale?
- 476. How do you configure distributed tracing from App Insights to SQL DB?
- 477. How do you baseline workload performance before scaling decisions?
- 478. How do you configure proactive alerts for deadlocks?
- 479. How do you monitor LTR backup storage costs over time?
- 480. How do you configure workload classification telemetry collection?

Hybrid & Multi-Cloud Scenarios

- 481. How do you integrate Azure SQL DB with on-prem Active Directory?
- 482. How do you replicate SQL DB data into AWS RDS for hybrid use cases?
- 483. How do you configure SQL DB as a source for multi-cloud analytics?
- 484. How do you configure data sync between Azure SQL DB and Google Cloud SQL?
- 485. How do you configure SQL DB failover strategy across clouds?
- 486. How do you secure hybrid SQL DB traffic over VPN tunnels?
- 487. How do you optimize latency in hybrid SQL DB applications?
- 488. How do you configure DNS resolution for hybrid SQL DB workloads?
- 489. How do you ensure consistent auditing across hybrid SQL environments?
- 490. How do you handle compliance across hybrid SQL workloads?

Future-Proofing & Best Practices

- 491. How do you prepare Azure SQL DB for future AI-driven workloads?
- 492. How do you configure Hyperscale tier for unpredictable growth?
- 493. How do you optimize for serverless adoption at scale?
- 494. How do you evaluate Business Critical vs Hyperscale trade-offs?
- 495. How do you configure Azure SQL DB for quantum-safe encryption readiness?
- 496. How do you enable automated index correction in long-term workloads?
- 497. How do you test new Azure SQL DB features in isolated sandboxes?
- 498. How do you build governance around SQL DB self-service provisioning?
- 499. How do you configure high availability in multi-region SaaS?
- 500. How do you enforce enterprise-wide standards for Azure SQL DB deployments?

Advanced (20)

- 481. How do you configure partitioned tables in SQL Server?
- 482. What is stretch database in SQL Server?
- 483. How do you configure PolyBase in SQL Server?
- 484. What is In-Memory OLTP and when to use it?
- 485. How do you monitor memory-optimized tables?
- 486. How do you configure Resource Governor workload groups?
- 487. How do you configure FILETABLE in SQL Server?
- 488. How do you configure full-text search?
- 489. What is the difference between columnstore index and rowstore index?
- 490. How do you configure clustered columnstore indexes?
- 491. How do you monitor columnstore index usage?
- 492. What are filtered statistics?
- 493. How do you configure SQL Server linked servers securely?
- 494. How do you configure OPENQUERY and pass-through queries?

495. What is the role of SQLCMD utility?
496. How do you automate SQL Server deployments with DACPAC?
497. How do you configure distributed transactions with MSDTC?
498. How do you configure SQL Server on Linux?
499. How do you manage SQL Server with containers?
500. What are SQL Server 2022 new features and improvements?

Real-Time AWS RDS & EC2 Administration Interview Questions

1. AWS EC2 – Instance Lifecycle & Provisioning (60)

1. How do you decide between On-Demand, Reserved, and Spot Instances?
2. How do you choose the right EC2 instance family for a workload?
3. How do you scale out EC2 instances during sudden traffic spikes?
4. What happens when you stop and start an EC2 instance with an instance store volume?
5. How do you handle EC2 instance hibernation?
6. How do you resize an EC2 instance type without downtime?
7. How do you recover from a failed EC2 launch due to capacity errors?
8. How do you move an EC2 instance to another subnet?
9. How do you troubleshoot EC2 instance launch failures?
10. How do you migrate an EC2 instance between regions?
11. What happens to EBS volumes when you terminate an instance?
12. How do you configure termination protection for EC2?
13. How do you capture a custom AMI from a running instance?
14. How do you troubleshoot when an instance fails status checks?
15. How do you recover an EC2 instance stuck in stopping state?
16. How do you configure EC2 launch templates?
17. What is the difference between launch templates and launch configurations?
18. How do you handle AMI deprecation in your environment?
19. How do you configure auto-recovery for failed EC2 instances?
20. How do you configure EC2 scheduled events?
21. What happens if you run out of Elastic IPs?
22. How do you use placement groups for high-performance workloads?
23. How do you configure cluster placement group for low-latency?
24. What is spread placement group and when to use it?
25. What is partition placement group and when to use it?
26. How do you troubleshoot EC2 capacity reservation issues?
27. How do you configure dedicated hosts vs dedicated instances?
28. How do you recover from accidental termination of an EC2 instance?
29. How do you monitor EC2 lifecycle events?
30. How do you share AMIs across accounts?
31. How do you encrypt AMIs for compliance?
32. How do you handle EC2 Spot Instance interruptions?
33. How do you configure EC2 Spot Fleets?
34. How do you configure EC2 capacity reservations for critical apps?
35. How do you use Savings Plans vs Reserved Instances for EC2?
36. How do you migrate workloads from On-Demand to Spot safely?
37. How do you troubleshoot failed AMI creation?
38. How do you configure EC2 launch with custom user data scripts?
39. How do you troubleshoot failed user data execution?

40. How do you move EC2 instance to a different VPC?
41. How do you upgrade EC2 instance OS version with minimal downtime?
42. How do you configure lifecycle hooks in Auto Scaling for EC2?
43. How do you patch EC2 instances automatically?
44. How do you configure AWS Systems Manager for EC2 management?
45. How do you handle EC2 drift detection?
46. How do you configure hybrid workloads with EC2 and on-premises servers?
47. How do you manage spot capacity pools?
48. How do you troubleshoot high reboot times in EC2?
49. How do you configure Golden AMI pipeline for EC2?
50. How do you manage EC2 inventories across multiple accounts?
51. How do you enforce tagging compliance for EC2 instances?
52. How do you enforce EC2 launch restrictions by IAM policy?
53. How do you configure EC2 resource quotas?
54. How do you troubleshoot root device volume corruption?
55. How do you boot an EC2 instance from a snapshot?
56. How do you recover EC2 instance with accidental OS-level misconfiguration?
57. How do you configure EC2 auto-healing with CloudWatch alarms?
58. How do you troubleshoot high boot time on Windows EC2?
59. How do you configure launch templates with mixed instance policies?
60. How do you troubleshoot EC2 launch template versioning issues?

2. AWS EC2 – Storage & EBS (40)

61. How do you increase EBS volume size without downtime?
62. How do you change EBS volume type from gp2 to gp3?
63. How do you recover from EBS volume stuck in “modifying” state?
64. How do you configure provisioned IOPS for database workloads?
65. How do you troubleshoot high latency on EBS volumes?
66. How do you restore data from EBS snapshots?
67. How do you automate EBS snapshot lifecycle policies?
68. How do you copy EBS snapshots across regions?
69. How do you share encrypted snapshots between accounts?
70. How do you troubleshoot failed EBS snapshot copy?
71. How do you configure multi-attach volumes in EBS?
72. How do you monitor EBS burst balance credits?
73. How do you handle root volume corruption in Linux EC2?
74. How do you attach and mount additional volumes to running EC2?
75. How do you encrypt existing unencrypted EBS volumes?
76. How do you configure Elastic Volumes feature for resizing?
77. How do you handle snapshot restore performance issues?
78. How do you troubleshoot EBS performance using CloudWatch metrics?
79. How do you configure RAID on EBS volumes?
80. How do you recover accidentally deleted snapshots?
81. How do you reduce EBS costs while keeping backups?
82. How do you configure DLM (Data Lifecycle Manager) for EBS?
83. How do you troubleshoot volume detachment failures?
84. How do you handle EBS volumes stuck in “deleting”?
85. How do you configure cross-account snapshot sharing securely?
86. How do you manage EBS volume tagging at scale?
87. How do you ensure snapshot compliance for regulated workloads?

88. How do you monitor IOPS usage against provisioned limits?
89. How do you troubleshoot I/O credit starvation in gp2 volumes?
90. How do you perform point-in-time restore with EBS snapshots?
91. How do you reduce snapshot restore time for large volumes?
92. How do you back up encrypted volumes with KMS keys?
93. How do you configure Fast Snapshot Restore for EBS?
94. How do you troubleshoot volume attach failures across AZs?
95. How do you monitor EBS costs per workload?
96. How do you configure AMIs with pre-attached EBS volumes?
97. How do you configure EBS performance for SAP workloads?
98. How do you troubleshoot slow boot volumes?
99. How do you enforce EBS encryption by default?
100. How do you recover deleted volumes from backups?

3. AWS EC2 – Networking & Security (40)

101. How do you assign Elastic IPs to EC2 instances?
102. How do you troubleshoot EC2 not reachable via SSH?
103. How do you configure security groups for least privilege?
104. How do you recover lost key pair access to EC2?
105. How do you configure IAM roles for EC2 securely?
106. How do you enforce IMDSv2 on all EC2 instances?
107. How do you configure VPC endpoints for EC2 traffic?
108. How do you configure NACLs vs Security Groups for EC2?
109. How do you troubleshoot intermittent connectivity to EC2?
110. How do you configure EC2 instances with multiple ENIs?
111. How do you troubleshoot EC2 public IP changes after stop/start?
112. How do you configure bastion hosts for EC2 access?
113. How do you monitor SSH brute force attempts?
114. How do you configure VPC Reachability Analyzer for EC2?
115. How do you enforce HTTPS traffic only for EC2 apps?
116. How do you configure EC2 instances with IPv6?
117. How do you monitor network throughput on EC2?
118. How do you troubleshoot packet loss between EC2 instances?
119. How do you configure DNS resolution for EC2?
120. How do you configure hybrid connectivity with EC2?
121. How do you configure EC2 placement in private subnets?
122. How do you secure RDP access to Windows EC2?
123. How do you configure Session Manager for EC2 access?
124. How do you troubleshoot EC2 connectivity through Transit Gateway?
125. How do you configure firewall rules with AWS Network Firewall?
126. How do you enforce SSL termination on EC2?
127. How do you configure EC2 instances in multiple AZs with failover?
128. How do you troubleshoot “host unreachable” errors?
129. How do you configure Elastic Network Adapter (ENA) for high throughput?
130. How do you enforce private IP-only communication?
131. How do you configure VPN access to private EC2 instances?
132. How do you troubleshoot asymmetric routing issues?
133. How do you configure EC2 in Outposts?
134. How do you audit security group rule changes?
135. How do you enforce NACL rules for compliance?

- 136. How do you configure firewall on EC2 OS level vs AWS SG?
- 137. How do you restrict EC2 access to specific IP ranges?
- 138. How do you use Security Hub to monitor EC2 misconfigurations?
- 139. How do you configure GuardDuty to detect EC2 compromise?
- 140. How do you investigate compromised EC2 instance?

4. AWS RDS – Provisioning & Configuration (50)

- 141. How do you decide between RDS and EC2-based database?
- 142. How do you select between RDS MySQL, PostgreSQL, SQL Server, and Aurora?
- 143. How do you configure Multi-AZ deployment in RDS?
- 144. How do you enable automated backups in RDS?
- 145. How do you configure RDS read replicas?
- 146. How do you promote a read replica to standalone?
- 147. How do you configure RDS storage auto-scaling?
- 148. How do you resize RDS instance class with minimal downtime?
- 149. How do you configure parameter groups in RDS?
- 150. How do you configure option groups in RDS?
- 151. How do you troubleshoot failed RDS instance creation?
- 152. How do you configure RDS performance insights?
- 153. How do you configure Enhanced Monitoring for RDS?
- 154. How do you configure IAM authentication for RDS?
- 155. How do you configure SSL/TLS for RDS connections?
- 156. How do you configure RDS in a private subnet?
- 157. How do you troubleshoot RDS connectivity failures?
- 158. How do you monitor RDS backup window conflicts?
- 159. How do you recover RDS from “storage-full” errors?
- 160. How do you configure RDS parameter group changes without reboot?
- 161. How do you automate RDS snapshots?
- 162. How do you copy RDS snapshots across regions?
- 163. How do you share RDS snapshots across accounts?
- 164. How do you recover deleted RDS instance?
- 165. How do you configure Aurora Serverless for unpredictable workloads?
- 166. How do you configure RDS with cross-region read replicas?
- 167. How do you troubleshoot failed read replica creation?
- 168. How do you configure RDS proxy for connection pooling?
- 169. How do you configure Aurora Global Database?
- 170. How do you configure encryption for RDS with KMS?
- 171. How do you migrate RDS instance between AZs?
- 172. How do you troubleshoot high replica lag?
- 173. How do you troubleshoot backup failures in RDS?
- 174. How do you configure deletion protection for RDS?
- 175. How do you troubleshoot slow RDS failover?
- 176. How do you configure RDS Performance Schema for MySQL?
- 177. How do you monitor RDS query performance?
- 178. How do you enable log exports to CloudWatch?
- 179. How do you configure RDS event notifications?
- 180. How do you configure RDS custom DNS endpoints?
- 181. How do you upgrade RDS engine versions with minimal downtime?
- 182. How do you test failover in RDS Multi-AZ setup?
- 183. How do you migrate on-premises DB to RDS?

184. How do you configure logical replication in RDS PostgreSQL?
185. How do you handle parameter group conflicts across environments?
186. How do you configure RDS reserved instances?
187. How do you configure Aurora parallel query for analytics?
188. How do you configure Aurora Backtrack feature?
189. How do you configure Aurora failover priority?
190. How do you troubleshoot failed Aurora cluster creation?

AWS RDS – Backup, Recovery & Maintenance (60)

191. How do you configure automated backups in RDS and set retention policies?
192. How do you restore an RDS instance to a point in time?
193. How do you recover a deleted RDS instance using automated backups?
194. How do you copy RDS snapshots to another region for DR?
195. How do you schedule RDS snapshots using Lambda?
196. How do you monitor snapshot storage costs?
197. How do you troubleshoot snapshot copy failures?
198. How do you configure backup windows to avoid performance impact?
199. How do you encrypt snapshots with a different KMS key?
200. How do you restore a specific database from a multi-DB RDS instance?
201. How do you recover corrupted RDS instance files?
202. How do you validate backups for consistency?
203. How do you configure RDS snapshots for compliance (e.g., HIPAA)?
204. How do you restore RDS instance to a different VPC?
205. How do you automate deletion of old RDS snapshots?
206. How do you troubleshoot slow restore of RDS snapshots?
207. How do you recover from accidental DROP TABLE in RDS?
208. How do you configure point-in-time restore for Aurora?
209. How do you recover failed Aurora cluster from backups?
210. How do you manage backup retention for Aurora Global Database?
211. How do you copy encrypted snapshots across accounts?
212. How do you test RDS recovery procedures regularly?
213. How do you configure cross-region backups for RDS SQL Server?
214. How do you recover system databases in RDS SQL Server?
215. How do you configure logical backups in RDS PostgreSQL?
216. How do you handle backup failures due to insufficient storage?
217. How do you restore a subset of databases from RDS SQL Server snapshot?
218. How do you troubleshoot automated backup failures?
219. How do you configure RDS snapshots lifecycle with AWS Backup?
220. How do you verify log backups in RDS SQL Server?
221. How do you export RDS snapshot to S3?
222. How do you configure Aurora Backtrack for near real-time restore?
223. How do you troubleshoot Aurora Backtrack limitations?
224. How do you configure incremental snapshot copies?
225. How do you restore RDS in another AWS account?
226. How do you monitor backup job metrics in CloudWatch?
227. How do you perform PITR (Point in Time Recovery) across regions?
228. How do you recover when backup window overlaps with maintenance window?
229. How do you configure Lambda triggers after snapshot creation?
230. How do you encrypt on-the-fly during snapshot copy?
231. How do you recover lost schema objects without full restore?

- 232. How do you configure snapshot sharing securely for partners?
- 233. How do you restore Aurora cluster to specific transaction ID?
- 234. How do you troubleshoot incomplete PITR restores?
- 235. How do you monitor backup failures with Event Subscriptions?
- 236. How do you implement PITR with MySQL binary logs in RDS?
- 237. How do you archive old RDS snapshots to Glacier?
- 238. How do you troubleshoot snapshot restore stuck in “creating”?
- 239. How do you manage snapshot IAM policies?
- 240. How do you ensure zero data loss in backups during failover?
- 241. How do you configure maintenance windows for critical RDS workloads?
- 242. How do you apply minor version upgrades with minimal downtime?
- 243. How do you defer maintenance actions for critical workloads?
- 244. How do you test RDS failover impact on backups?
- 245. How do you configure multi-account backup compliance?
- 246. How do you perform selective restore of schema objects?
- 247. How do you export RDS SQL Server backups to native BAK files?
- 248. How do you recover RDS instance from AWS Backup vault?
- 249. How do you automate tagging of RDS snapshots?
- 250. How do you test RDS recovery time objectives (RTO)?

6. AWS RDS – Performance Tuning & Monitoring (60)

- 251. How do you monitor CPU, memory, and storage utilization in RDS?
- 252. How do you troubleshoot high CPU in RDS PostgreSQL?
- 253. How do you identify slow queries in RDS MySQL?
- 254. How do you tune RDS SQL Server tempdb configuration?
- 255. How do you configure Performance Insights for workload analysis?
- 256. How do you troubleshoot replica lag in RDS read replicas?
- 257. How do you monitor IOPS consumption in RDS?
- 258. How do you troubleshoot query timeouts in RDS?
- 259. How do you monitor and tune max connections in RDS?
- 260. How do you analyze query execution plans in RDS PostgreSQL?
- 261. How do you detect deadlocks in RDS SQL Server?
- 262. How do you troubleshoot disk space growth in RDS?
- 263. How do you configure Enhanced Monitoring with CloudWatch Logs?
- 264. How do you tune autovacuum in RDS PostgreSQL?
- 265. How do you troubleshoot high replication lag in Aurora?
- 266. How do you monitor database connections using CloudWatch metrics?
- 267. How do you troubleshoot query cache misses in MySQL RDS?
- 268. How do you configure alarms for freeable memory in RDS?
- 269. How do you detect index fragmentation in RDS SQL Server?
- 270. How do you tune RDS PostgreSQL work_mem and shared_buffers?
- 271. How do you troubleshoot Aurora performance regression after failover?
- 272. How do you identify top queries in RDS using Performance Insights?
- 273. How do you troubleshoot connection pool exhaustion?
- 274. How do you detect locking/blocking issues in RDS?
- 275. How do you monitor RDS transaction log usage?
- 276. How do you tune checkpoint settings in PostgreSQL RDS?
- 277. How do you troubleshoot IO wait events in Aurora?
- 278. How do you configure alarms for burst balance credits?
- 279. How do you analyze wait events in RDS PostgreSQL?

280. How do you tune query parallelism in Aurora PostgreSQL?
281. How do you configure monitoring for SQL Server Agent jobs in RDS?
282. How do you troubleshoot CPU throttling in burstable RDS instances?
283. How do you monitor RDS SQL Server deadlock graphs?
284. How do you detect inefficient queries in RDS?
285. How do you configure auto-scaling storage alerts?
286. How do you tune redo log settings in RDS Oracle?
287. How do you troubleshoot high tempdb usage in RDS SQL Server?
288. How do you monitor cache hit ratios in RDS PostgreSQL?
289. How do you detect long-running queries in Aurora MySQL?
290. How do you optimize Aurora writer and reader scaling?
291. How do you analyze log files exported to CloudWatch?
292. How do you detect resource contention in RDS Multi-AZ setup?
293. How do you configure event subscriptions for performance alerts?
294. How do you troubleshoot frequent connection drops in RDS?
295. How do you identify inefficient indexes in RDS MySQL?
296. How do you configure metrics collection using CloudWatch Agent?
297. How do you troubleshoot high CPU in Aurora PostgreSQL?
298. How do you detect excessive autovacuum activity in PostgreSQL RDS?
299. How do you troubleshoot IO contention in gp2/gp3 storage?
300. How do you configure proactive query optimization with Advisor?
301. How do you handle max IOPS scaling limitations?
302. How do you troubleshoot buffer cache pressure in SQL Server RDS?
303. How do you use pg_stat_statements for query tuning?
304. How do you monitor Aurora Global Database replication lag?
305. How do you analyze workload with Query Store in RDS SQL Server?
306. How do you troubleshoot Aurora cluster endpoint latency?
307. How do you configure CloudWatch dashboards for RDS metrics?
308. How do you tune Aurora parallel query performance?
309. How do you detect lock contention patterns?
310. How do you configure alarms for RDS free storage space?

7. AWS RDS – High Availability & DR (40)

311. How do you configure RDS Multi-AZ for failover protection?
312. How do you test failover scenarios in RDS?
313. How do you configure RDS read replicas for HA?
314. How do you configure Aurora Global Database for DR?
315. How do you troubleshoot slow RDS failover?
316. How do you design cross-region replication for compliance?
317. How do you configure DNS failover with Route 53 for RDS?
318. How do you perform manual failover in RDS?
319. How do you test Aurora Global Database recovery?
320. How do you configure cross-account DR strategy for RDS?
321. How do you troubleshoot replication failures in RDS MySQL?
322. How do you automate DR testing for RDS?
323. How do you monitor failover events in RDS?
324. How do you simulate regional outage for RDS?
325. How do you configure active-active replication with Aurora?
326. How do you recover from Aurora cluster quorum loss?
327. How do you configure RDS Proxy for failover handling?

328. How do you configure backup-based DR strategy?
329. How do you troubleshoot DNS propagation delays after failover?
330. How do you handle failover during peak workloads?
331. How do you configure automatic failover priority in Aurora?
332. How do you recover from stuck failover state?
333. How do you handle maintenance failovers proactively?
334. How do you implement warm standby for RDS SQL Server?
335. How do you troubleshoot inconsistent replication lag?
336. How do you monitor HA health metrics in RDS?
337. How do you configure Route 53 health checks for RDS endpoints?
338. How do you perform cross-region failover manually?
339. How do you simulate AZ outage to test HA?
340. How do you ensure zero data loss in Aurora DR?
341. How do you design RPO/RTO objectives with RDS?
342. How do you troubleshoot failover delays in Aurora PostgreSQL?
343. How do you monitor HA events with Event Subscriptions?
344. How do you recover from replication slot bloat in PostgreSQL RDS?
345. How do you configure HA testing in non-production environments?
346. How do you handle failover with client connection pools?
347. How do you configure hybrid DR between RDS and on-premises DB?
348. How do you ensure replication encryption in cross-region setup?
349. How do you simulate large-scale DR event for compliance testing?
350. How do you automate DR recovery playbooks for RDS?

8. AWS EC2 – Monitoring, Scaling & Troubleshooting (60)

351. How do you monitor EC2 instance metrics using CloudWatch?
352. How do you troubleshoot high CPU in an EC2 instance?
353. How do you investigate memory leaks in EC2?
354. How do you analyze disk space issues in EC2?
355. How do you troubleshoot slow boot times in EC2?
356. How do you configure CloudWatch Agent for custom metrics?
357. How do you use Systems Manager to collect instance logs?
358. How do you troubleshoot application crashes on EC2?
359. How do you monitor processes inside EC2 automatically?
360. How do you configure alarms for EC2 instance health?
361. How do you troubleshoot high network latency in EC2?
362. How do you configure auto-scaling policies for EC2 groups?
363. How do you tune scaling cooldown timers?
364. How do you troubleshoot failed scale-out events?
365. How do you configure predictive scaling policies?
366. How do you test scaling policies under load?
367. How do you configure scheduled scaling for EC2?
368. How do you monitor cost impact of scaling events?
369. How do you troubleshoot scaling failures due to resource limits?
370. How do you configure target tracking scaling policies?
371. How do you analyze Auto Scaling activity logs?
372. How do you troubleshoot launch configuration mismatches?
373. How do you configure dynamic scaling with CloudWatch metrics?
374. How do you monitor unhealthy EC2 instances in Auto Scaling?
375. How do you handle failed EC2 instance replacement?

376. How do you troubleshoot Auto Scaling lifecycle hook failures?
377. How do you analyze root cause of scaling flapping?
378. How do you test Auto Scaling rollback scenarios?
379. How do you configure mixed instance types in scaling groups?
380. How do you troubleshoot scaling issues in Spot instances?
381. How do you monitor horizontal scaling effectiveness?
382. How do you configure Elastic Load Balancer with Auto Scaling?
383. How do you troubleshoot ELB health check failures?
384. How do you analyze traffic distribution across EC2 in ELB?
385. How do you configure cross-zone load balancing?
386. How do you troubleshoot session stickiness issues in ELB?
387. How do you configure SSL termination on ELB with EC2 backend?
388. How do you monitor ELB latency metrics?
389. How do you troubleshoot failed ELB connections?
390. How do you configure ALB routing rules for EC2 targets?
391. How do you troubleshoot target registration failures in ALB?
392. How do you test failover between AZs with Auto Scaling?
393. How do you configure CloudWatch anomaly detection for EC2 metrics?
394. How do you troubleshoot excessive instance replacements?
395. How do you monitor scaling policies with AWS Trusted Advisor?
396. How do you troubleshoot IAM policy restrictions for Auto Scaling?
397. How do you perform load testing for Auto Scaling validation?
398. How do you configure hybrid scaling (EC2 + ECS/EKS)?
399. How do you troubleshoot orphaned instances in Auto Scaling?
400. How do you configure graceful shutdown during scale-in?
401. How do you configure Lambda for custom Auto Scaling actions?
402. How do you monitor scaling latency for time-sensitive apps?
403. How do you troubleshoot insufficient capacity for scale-out?
404. How do you configure predictive scaling for batch workloads?
405. How do you troubleshoot Auto Scaling warm pools?
406. How do you tune health check grace periods?
407. How do you configure Spot Fleets with Auto Scaling groups?
408. How do you monitor cost optimization in scaling policies?
409. How do you troubleshoot over-provisioning due to scaling misconfiguration?
410. How do you configure failover between Spot and On-Demand instances?

9. AWS EC2 & RDS – Security, Compliance & Cost Optimization (60)

411. How do you enforce encryption for RDS instances?
412. How do you configure IAM database authentication in RDS?
413. How do you audit RDS logins and queries?
414. How do you configure CloudTrail for EC2/RDS activity?
415. How do you detect unauthorized login attempts to RDS?
416. How do you enforce SSL connections to RDS?
417. How do you configure VPC peering for secure DB access?
418. How do you enforce IAM roles for EC2 least privilege?
419. How do you detect unused RDS instances?
420. How do you configure Security Hub to monitor RDS compliance?
421. How do you audit encryption keys used in RDS?
422. How do you enforce encryption at rest for EC2 volumes?
423. How do you detect unencrypted EBS volumes automatically?

424. How do you configure GuardDuty to detect EC2/RDS anomalies?
425. How do you secure RDP access to Windows EC2 at scale?
426. How do you enforce password policies in RDS?
427. How do you monitor RDS user activity with CloudWatch logs?
428. How do you rotate RDS master user passwords automatically?
429. How do you configure Secrets Manager for DB credentials?
430. How do you troubleshoot KMS key access errors in RDS?
431. How do you enforce restricted ports in EC2 security groups?
432. How do you automate patching for RDS instances?
433. How do you configure Inspector to scan EC2 vulnerabilities?
434. How do you analyze RDS logs for suspicious activity?
435. How do you configure IAM policies for cross-account EC2/RDS access?
436. How do you secure EC2 user data scripts from secrets leakage?
437. How do you configure audit logging for SQL Server RDS?
438. How do you enforce encryption for DB connections end-to-end?
439. How do you monitor failed login attempts on EC2 Linux instances?
440. How do you enforce private subnets for RDS?
441. How do you use VPC endpoints for RDS connectivity?
442. How do you perform compliance audits for EC2 instances?
443. How do you restrict EC2 instance launch to specific AMIs?
444. How do you enforce mandatory tagging for EC2/RDS?
445. How do you configure SCPs (Service Control Policies) for RDS usage?
446. How do you detect idle EC2 instances?
447. How do you configure budgets for RDS/EC2 costs?
448. How do you analyze cost breakdown of EC2 storage vs compute?
449. How do you configure Compute Optimizer for EC2 rightsizing?
450. How do you troubleshoot sudden cost spikes in RDS?
451. How do you implement cost anomaly detection for RDS?
452. How do you use Reserved Instances for RDS cost savings?
453. How do you evaluate Savings Plans vs RI for RDS workloads?
454. How do you configure Spot Instances for cost optimization?
455. How do you monitor cross-region data transfer costs for RDS?
456. How do you enforce tagging-based cost allocation?
457. How do you troubleshoot RDS over-provisioning issues?
458. How do you configure Trusted Advisor checks for EC2/RDS savings?
459. How do you enforce limits on RDS storage auto-scaling?
460. How do you monitor idle connections cost in Aurora Serverless?
461. How do you migrate on-premises SQL Server to RDS SQL Server?
462. How do you migrate Oracle DB to RDS PostgreSQL?

RDS Advanced Performance & Troubleshooting (463–480)

463. How do you troubleshoot sudden connection spikes on an RDS PostgreSQL instance?
464. How do you detect and mitigate table bloat in RDS PostgreSQL without superuser access?
465. How do you handle RDS SQL Server blocking and deadlocks?
466. How do you troubleshoot long-running queries without OS-level access?
467. How do you optimize RDS Aurora for mixed OLTP + reporting workloads?
468. How do you identify queries causing high CPU in RDS MySQL?
469. How do you enable and analyze slow query logs in RDS?
470. How do you troubleshoot “RDS storage-optimized burst credits depleted” issue?
471. How do you reduce RDS IOPS costs without hurting performance?

472. How do you troubleshoot write latency in Multi-AZ RDS setup?
473. How do you troubleshoot replication failures in RDS SQL Server?
474. How do you investigate increased RDS Aurora replica lag?
475. How do you troubleshoot RDS PostgreSQL autovacuum not running?
476. How do you monitor and resolve transaction ID wraparound issues in RDS PostgreSQL?
477. How do you tune RDS SQL Server tempdb for high-concurrency workloads?
478. How do you troubleshoot sudden failover events triggered in RDS?
479. How do you minimize downtime during unexpected RDS storage scaling?
480. How do you fix connection timeouts when RDS is inside a private subnet?

RDS Migration & DR (481–490)

481. How do you migrate from on-prem SQL Server to RDS with near-zero downtime?
482. How do you handle schema conversion challenges during RDS migration?
483. How do you test RDS failover in a DR drill without impacting production?
484. How do you configure cross-region RDS read replica as a DR strategy?
485. How do you handle RDS migration from Oracle to PostgreSQL?
486. How do you configure AWS DMS for continuous replication into RDS?
487. How do you reduce downtime when migrating a 10TB database into RDS?
488. How do you recover a corrupted RDS snapshot during migration testing?
489. How do you design an RDS multi-region active/standby architecture?
490. How do you test point-in-time recovery in RDS for compliance audits?

EC2 Advanced Performance & Cost Optimization (491–500)

491. How do you troubleshoot EC2 application slowness caused by CPU throttling?
492. How do you identify “noisy neighbor” issues on shared tenancy EC2?
493. How do you optimize EC2 cost when workloads have predictable schedules?
494. How do you configure EC2 Spot Instances for fault-tolerant workloads?
495. How do you troubleshoot an EC2 instance stuck at 100% network throughput?
496. How do you use Compute Optimizer to right-size EC2 instances?
497. How do you handle EC2 auto-scaling during sudden 5x traffic surges?
498. How do you configure EC2 with multiple EBS volumes for performance (striping)?
499. How do you enforce security baseline on EC2 instances across 100+ accounts?
500. How do you design an EC2 + RDS architecture to balance performance and cost for a SaaS workload?

Real-Time Scenarios Interview Questions on Azure SQL Database Administration

Provisioning & Configuration

1. How do you decide between Azure SQL Database single database and elastic pool?
2. How do you choose between DTU-based and vCore-based purchasing models?
3. How do you select the right service tier (General Purpose, Business Critical, Hyperscale)?
4. How do you configure Azure SQL DB for unpredictable workloads?
5. How do you migrate from DTU-based to vCore-based model with minimal downtime?
6. How do you enable zone redundancy for Business Critical tier?
7. How do you resize Azure SQL Database compute without downtime?
8. How do you configure Azure SQL Hyperscale for large databases?
9. How do you provision a serverless Azure SQL DB for intermittent workloads?
10. How do you set up auto-pause and auto-resume in serverless databases?
11. How do you configure Azure SQL DB for multi-tenant SaaS architecture?
12. How do you configure geo-redundant backup storage for Azure SQL DB?

13. How do you manage connection strings securely in Azure SQL DB?
14. How do you assign a custom DNS name for your SQL logical server?
15. How do you enable availability zone support in Azure SQL DB?
16. How do you configure Azure SQL DB to comply with GDPR data residency requirements?
17. How do you configure Azure SQL DB deployment across multiple regions?
18. How do you configure elastic pools for cost efficiency?
19. How do you move a database in and out of an elastic pool?
20. How do you scale storage independently of compute in Hyperscale?

Security

21. How do you configure Transparent Data Encryption (TDE) in Azure SQL DB?
22. How do you implement customer-managed keys for TDE?
23. How do you enforce Always Encrypted in Azure SQL DB?
24. How do you configure Azure AD authentication for Azure SQL DB?
25. How do you configure conditional access for SQL Database administrators?
26. How do you implement Row-Level Security (RLS)?
27. How do you configure Dynamic Data Masking (DDM)?
28. How do you enforce TLS 1.2 connections only?
29. How do you configure auditing for Azure SQL DB?
30. How do you store audit logs in a Log Analytics workspace?
31. How do you detect SQL injection attempts in Azure SQL DB?
32. How do you configure firewall rules for Azure SQL DB at server and database level?
33. How do you configure VNet service endpoints for Azure SQL DB?
34. How do you configure Private Link for Azure SQL DB?
35. How do you enforce "no public access" for Azure SQL DB?
36. How do you enable Azure Defender for SQL?
37. How do you review security alerts from Azure Defender?
38. How do you mitigate brute-force login attempts against Azure SQL DB?
39. How do you configure Just-In-Time (JIT) access for administrators?
40. How do you enable and monitor SQL Vulnerability Assessments?

High Availability & Disaster Recovery

41. How do you configure Active Geo-Replication for Azure SQL DB?
42. How do you test failover for geo-replicated databases?
43. How do you monitor replication lag in active geo-replication?
44. How do you configure Auto-Failover Groups across regions?
45. How do you simulate regional failover in Auto-Failover Groups?
46. How do you ensure DNS endpoint cutover in failover groups?
47. How do you configure zone redundancy for Business Critical databases?
48. How do you configure point-in-time restore for a deleted database?
49. How do you recover from accidental table deletion in Azure SQL DB?
50. How do you configure long-term retention (LTR) backups?
51. How do you restore from LTR backup into a different server?
52. How do you automate DR testing for Azure SQL DB?
53. How do you minimize downtime during failover testing?
54. How do you handle failover with Hyperscale tier databases?
55. How do you configure backup storage redundancy (LRS, GRS, ZRS)?
56. How do you monitor backup storage usage?
57. How do you validate restore capability from backups?
58. How do you configure cross-subscription DR for Azure SQL DB?

59. How do you manage RPO and RTO objectives in Azure SQL DB?
60. How do you configure backup policies to meet compliance standards?

Performance Tuning

61. How do you troubleshoot high DTU usage in a database?
62. How do you identify top resource-consuming queries?
63. How do you use Query Performance Insight?
64. How do you use Intelligent Insights for performance diagnostics?
65. How do you use Automatic Tuning to fix performance regressions?
66. How do you force plan corrections in Azure SQL DB?
67. How do you identify blocking sessions?
68. How do you troubleshoot deadlocks in Azure SQL DB?
69. How do you capture deadlock graphs from Azure SQL DB?
70. How do you tune queries in Hyperscale databases?
71. How do you analyze wait statistics in Azure SQL DB?
72. How do you monitor IO utilization in Azure SQL DB?
73. How do you detect tempdb contention?
74. How do you handle parameter sniffing issues?
75. How do you use Extended Events in Azure SQL DB?
76. How do you capture slow queries with Query Store?
77. How do you analyze query store performance regressions?
78. How do you manage query store size and cleanup?
79. How do you force last known good plan using Query Store?
80. How do you troubleshoot excessive log writes?

Monitoring & Alerts

81. How do you configure alerts for high DTU usage?
82. How do you configure alerts for high CPU usage?
83. How do you configure alerts for deadlocks?
84. How do you integrate Azure SQL metrics with Azure Monitor?
85. How do you push Azure SQL logs to Log Analytics?
86. How do you configure alerts for storage consumption?
87. How do you detect login failures?
88. How do you configure custom dashboards for SQL performance?
89. How do you monitor connections per database?
90. How do you monitor blocking sessions via alerts?
91. How do you configure activity log alerts for SQL changes?
92. How do you monitor geo-replication health?
93. How do you configure alerts for failover events?
94. How do you detect auto-tuning recommendations?
95. How do you configure proactive tuning alerts?
96. How do you monitor query store performance?
97. How do you detect throttling events in Azure SQL DB?
98. How do you integrate Azure SQL DB monitoring with Grafana?
99. How do you use Azure SQL Analytics for fleet-wide monitoring?
100. How do you configure workload classification for monitoring?

Migrations

101. How do you migrate on-prem SQL Server to Azure SQL DB with minimal downtime?
102. How do you use DMA (Data Migration Assistant) for assessment?
103. How do you use DMS (Database Migration Service) for online migration?
104. How do you handle unsupported features in Azure SQL DB during migration?
105. How do you migrate from Azure SQL Managed Instance to Azure SQL DB?
106. How do you migrate from AWS RDS SQL Server to Azure SQL DB?
107. How do you migrate a 5TB database to Hyperscale tier?
108. How do you migrate SQL Server Agent jobs during migration?
109. How do you migrate linked servers to Azure SQL DB?
110. How do you handle cross-database queries after migration?
111. How do you test application compatibility post-migration?
112. How do you migrate users and logins to Azure SQL DB?
113. How do you migrate SSIS packages into Azure Data Factory?
114. How do you handle CLR objects in migration?
115. How do you migrate SQL Server Reporting Services (SSRS) to Azure?
116. How do you migrate partitioned tables?
117. How do you migrate Always Encrypted data?
118. How do you migrate a large transactional workload with near-zero downtime?
119. How do you validate data consistency after migration?
120. How do you rollback migration if Azure SQL DB performance is unsatisfactory?

Automation & DevOps

121. How do you automate database deployment with ARM templates?
122. How do you provision Azure SQL DB using Terraform?
123. How do you configure CI/CD pipelines with Azure DevOps for SQL DB?
124. How do you deploy schema changes using DACPACs?
125. How do you automate data masking in non-prod environments?
126. How do you schedule database maintenance tasks without SQL Agent?
127. How do you implement jobs with Elastic Jobs in Azure SQL DB?
128. How do you configure PowerShell automation for backups?
129. How do you configure automatic index management with T-SQL and Azure Automation?
130. How do you automate scaling of Azure SQL DB based on metrics?
131. How do you integrate Azure SQL with GitHub Actions?
132. How do you enforce policy compliance using Azure Policy?
133. How do you configure drift detection with Azure DevOps?
134. How do you implement Infrastructure as Code for Azure SQL DB?
135. How do you configure schema drift alerts?
136. How do you set up automated performance baselines?
137. How do you implement data refresh pipelines from prod to dev/test?
138. How do you integrate Azure SQL with Bicep templates?
139. How do you configure automation for long-term retention cleanup?
140. How do you test automated failover with pipelines?

Cost Optimization

141. How do you decide between single database and elastic pool for cost savings?
142. How do you monitor Azure SQL costs per database?
143. How do you configure alerts for cost overruns?
144. How do you reduce cost using reserved capacity?
145. How do you optimize cost by moving from provisioned to serverless tier?

146. How do you identify unused or idle databases?
147. How do you scale down workloads during off-hours automatically?
148. How do you reduce backup storage costs in Azure SQL DB?
149. How do you optimize cost for Hyperscale databases?
150. How do you simulate cost impact of scaling up Azure SQL DB?
151. How do you consolidate workloads into elastic pools for savings?
152. How do you measure DTU vs vCore costs for performance tradeoffs?
153. How do you configure tagging for cost governance?
154. How do you analyze cost by application or department?
155. How do you reduce cost with auto-pause in serverless DBs?
156. How do you configure alerts for unused elastic pool capacity?
157. How do you right-size vCore allocation in Business Critical workloads?
158. How do you use Azure Advisor for SQL cost optimization?
159. How do you calculate long-term savings with reserved instances?
160. How do you design cost governance strategy across subscriptions?
161. How do you automate index rebuilds in Azure SQL DB without SQL Agent?
162. How do you decide between index rebuild vs reorganize in cloud databases?
163. How do you monitor index fragmentation in Azure SQL DB?
164. How do you optimize index maintenance for large tables in Hyperscale?
165. How do you use automatic index management in Azure SQL DB?
166. How do you troubleshoot auto-created indexes that degrade performance?
167. How do you monitor unused indexes in Azure SQL DB?
168. How do you detect missing index recommendations from Query Store?
169. How do you handle large partitioned index maintenance in Azure SQL DB?
170. How do you automate statistics updates in Azure SQL DB?
171. How do you validate if statistics are stale in serverless workloads?
172. How do you monitor statistics update impact on query performance?
173. How do you force full scan stats update for skewed data distribution?
174. How do you configure maintenance with Azure Automation Runbooks?
175. How do you optimize fill factor settings for indexes in Azure SQL DB?
176. How do you handle maintenance jobs across 100+ Azure SQL DBs?
177. How do you test index impact before applying recommendations?
178. How do you schedule maintenance using Elastic Jobs?
179. How do you prevent auto-drop of indexes by auto-tuning?
180. How do you measure impact of statistics auto-update on large tables?

Scalability & Elasticity

181. How do you scale Azure SQL DB up during traffic spikes?
182. How do you configure autoscale rules with serverless tier?
183. How do you detect when to scale out using read replicas?
184. How do you use elastic pools for unpredictable workloads?
185. How do you configure sharding in Azure SQL DB?
186. How do you implement horizontal partitioning for SaaS applications?
187. How do you handle connection distribution across shards?
188. How do you configure Azure SQL Hyperscale read replicas?
189. How do you optimize data distribution in elastic pools?
190. How do you configure database per tenant strategy in Azure SQL DB?
191. How do you monitor connection exhaustion in elastic pools?
192. How do you configure connection resiliency with retry logic?
193. How do you reduce latency in globally distributed Azure SQL DB?

- 194. How do you design cross-region read scalability?
- 195. How do you configure caching layers in front of Azure SQL DB?
- 196. How do you optimize queries in multi-tenant elastic pool environments?
- 197. How do you configure shard map manager in Azure SQL DB?
- 198. How do you scale storage in Hyperscale tier dynamically?
- 199. How do you balance performance and cost in elastic pools?
- 200. How do you monitor scaling events in Azure SQL DB?

Compliance & Auditing

- 201. How do you configure audit logs for compliance frameworks like HIPAA?
- 202. How do you monitor unauthorized schema changes in Azure SQL DB?
- 203. How do you configure retention policies for audit logs?
- 204. How do you ensure audit logs cannot be tampered with?
- 205. How do you integrate SQL audit logs into SIEM tools?
- 206. How do you configure Azure Policy to enforce encryption at rest?
- 207. How do you configure Azure Monitor to alert on unauthorized logins?
- 208. How do you prove encryption compliance in Azure SQL DB to auditors?
- 209. How do you configure periodic access reviews for SQL admins?
- 210. How do you detect privilege escalations in Azure SQL DB?
- 211. How do you restrict access to production data in dev/test?
- 212. How do you configure masking rules for compliance exports?
- 213. How do you configure long-term retention for compliance data?
- 214. How do you configure immutable logs for forensic investigations?
- 215. How do you ensure compliance with PCI DSS in Azure SQL DB?
- 216. How do you validate audit completeness during DR drills?
- 217. How do you export logs securely to external auditors?
- 218. How do you enforce least-privilege access with managed identities?
- 219. How do you detect dormant user accounts in Azure SQL DB?
- 220. How do you configure custom auditing filters for sensitive tables?

Connectivity & Networking

- 221. How do you configure Private Endpoints for Azure SQL DB?
- 222. How do you configure VNet integration for Azure SQL DB?
- 223. How do you troubleshoot connectivity from on-prem to Azure SQL DB?
- 224. How do you configure ExpressRoute for Azure SQL DB connectivity?
- 225. How do you enforce traffic over Private Link only?
- 226. How do you configure firewall rules for hybrid connectivity?
- 227. How do you troubleshoot intermittent connectivity issues?
- 228. How do you test connectivity from an Azure VM to SQL DB?
- 229. How do you configure routing for SQL DB in secured VNets?
- 230. How do you integrate SQL DB with Application Gateway?
- 231. How do you configure VPN access to Azure SQL DB?
- 232. How do you monitor failed login attempts from external IPs?
- 233. How do you restrict connectivity to specific Azure services?
- 234. How do you enable service endpoint policies for SQL DB?
- 235. How do you monitor egress traffic from SQL DB to detect anomalies?
- 236. How do you configure TLS enforcement for all connections?
- 237. How do you rotate certificates for TLS in SQL DB?
- 238. How do you troubleshoot SSL handshake failures?
- 239. How do you test failover group endpoint connectivity?

240. How do you configure DNS resolution for SQL DB across VNets?

Advanced Features & Capabilities

- 241. How do you configure in-memory OLTP in Azure SQL DB?
- 242. How do you use temporal tables for auditing changes?
- 243. How do you use graph tables in Azure SQL DB?
- 244. How do you use columnstore indexes for analytics workloads?
- 245. How do you configure partitioned tables in Azure SQL DB?
- 246. How do you optimize performance of partitioned tables?
- 247. How do you configure stretch database for hybrid scenarios?
- 248. How do you configure JSON storage and querying in SQL DB?
- 249. How do you configure full-text search in Azure SQL DB?
- 250. How do you configure spatial data types in SQL DB?
- 251. How do you configure PolyBase for external data sources?
- 252. How do you integrate SQL DB with Azure Synapse for analytics?
- 253. How do you configure materialized views for performance?
- 254. How do you optimize batch workloads in Azure SQL DB?
- 255. How do you configure change tracking vs CDC in SQL DB?
- 256. How do you capture DML changes in Azure SQL DB without CDC?
- 257. How do you configure distributed transactions across SQL DBs?
- 258. How do you configure service broker equivalents in SQL DB?
- 259. How do you implement queue-based architectures in SQL DB?
- 260. How do you configure CLR alternatives in Azure SQL DB?

Disaster Scenarios

- 261. How do you recover from accidental DROP DATABASE in Azure SQL DB?
- 262. How do you restore to a specific point in time before corruption?
- 263. How do you handle ransomware attacks targeting SQL DB?
- 264. How do you verify backups are restorable during audits?
- 265. How do you configure geo-replication to secondary region before hurricane risk?
- 266. How do you recover from geo-secondary failure?
- 267. How do you re-seed geo-replication after corruption detected?
- 268. How do you recover after Azure region outage?
- 269. How do you validate DR failback to primary region?
- 270. How do you restore deleted LTR backups?
- 271. How do you recover from transaction log growth exhausting space?
- 272. How do you troubleshoot database stuck in recovery mode?
- 273. How do you recover corrupted system tables in Azure SQL DB?
- 274. How do you handle schema corruption in production SQL DB?
- 275. How do you test time-to-recovery in failover group scenarios?
- 276. How do you simulate accidental schema drop recovery?
- 277. How do you configure playbooks for SQL DB outages?
- 278. How do you run DR drills across subscriptions?
- 279. How do you recover tenant database in multi-tenant SaaS?
- 280. How do you restore across subscriptions when prod server is lost?

Monitoring Large Fleets

281. How do you manage 500+ Azure SQL DBs across multiple subscriptions?
282. How do you configure central monitoring for all SQL DBs?
283. How do you configure Azure SQL Analytics at scale?
284. How do you configure budget alerts for large SQL fleets?
285. How do you enforce tagging compliance for all SQL DBs?
286. How do you monitor performance for all elastic pools centrally?
287. How do you manage user access at scale across many databases?
288. How do you roll out schema changes across hundreds of DBs?
289. How do you enforce security baselines across SQL DB fleet?
290. How do you implement centralized auditing solution?
291. How do you detect outlier performance in fleet-wide metrics?
292. How do you optimize cost distribution across business units?
293. How do you configure fleet-wide patch automation?
294. How do you configure workload isolation in elastic pools?
295. How do you set up proactive scaling rules fleet-wide?
296. How do you configure alerts for all SQL DBs centrally?
297. How do you detect compliance drift across many environments?
298. How do you manage LTR backups across multiple workloads?
299. How do you implement RBAC at scale for SQL DB admins?
300. How do you report consolidated SLA compliance for SQL DB?

Troubleshooting & Real-Time Issues

301. How do you troubleshoot queries suddenly running slow?
302. How do you investigate sudden DTU spikes at midnight?
303. How do you troubleshoot failed logins from Azure App Service?
304. How do you troubleshoot slow connectivity from hybrid apps?
305. How do you troubleshoot high tempdb usage in SQL DB?
306. How do you investigate "session terminated due to excessive tempdb"?
307. How do you troubleshoot transaction log growth in elastic pools?
308. How do you troubleshoot memory grants in large queries?
309. How do you troubleshoot row version store growth in tempdb?
310. How do you troubleshoot frequent deadlocks across tenants?
311. How do you troubleshoot blocking chains in Azure SQL DB?
312. How do you troubleshoot query store bloating database size?
313. How do you troubleshoot backups failing due to throttling?
314. How do you troubleshoot SSL/TLS connection errors?
315. How do you troubleshoot failed geo-replica seeding?
316. How do you troubleshoot failover group not failing over?
317. How do you troubleshoot excessive login failures from specific IP?
318. How do you troubleshoot Azure SQL DB not accessible after firewall change?
319. How do you troubleshoot application timeouts under load?
320. How do you troubleshoot incorrect plan forcing by auto-tuning?

Integrations & Ecosystem

321. How do you integrate Azure SQL DB with Azure Functions?
322. How do you configure SQL DB as a source for Power BI?
323. How do you integrate SQL DB with Azure Data Factory pipelines?
324. How do you configure Event Grid integration with SQL DB?
325. How do you configure Logic Apps for SQL DB monitoring?

- 326. How do you configure Change Tracking with Event Hubs for CDC?
- 327. How do you connect SQL DB with Synapse Link?
- 328. How do you integrate SQL DB with Databricks?
- 329. How do you configure SQL DB as linked service in Synapse pipelines?
- 330. How do you configure SQL DB for Stream Analytics workloads?
- 331. How do you integrate SQL DB with Azure Monitor Logs?
- 332. How do you configure Application Insights to monitor SQL queries?
- 333. How do you secure SQL DB connections from Azure Kubernetes Service?
- 334. How do you integrate SQL DB with Key Vault secrets?
- 335. How do you configure Managed Identity access for SQL DB apps?
- 336. How do you configure SQL DB as backend for API Management?
- 337. How do you configure hybrid queries using PolyBase with on-prem SQL?
- 338. How do you configure SQL DB alerts into Teams/Slack channels?
- 339. How do you integrate SQL DB audit logs with Microsoft Sentinel?
- 340. How do you configure SQL DB for event-driven architectures?

Advanced Security Scenarios

- 341. How do you rotate encryption keys in SQL DB without downtime?
- 342. How do you monitor failed Always Encrypted queries?
- 343. How do you configure certificate rotation for Always Encrypted?
- 344. How do you detect access to sensitive columns using auditing?
- 345. How do you configure automatic masking for production refreshes?
- 346. How do you configure row filters for multi-tenant SaaS security?
- 347. How do you configure Just-In-Time access for emergency troubleshooting?
- 348. How do you detect insider threats in SQL DB?
- 349. How do you enforce Azure AD-only authentication in SQL DB?
- 350. How do you configure conditional MFA for SQL admins?
- 351. How do you detect and block anomalous logins using Defender?
- 352. How do you enforce password complexity in contained users?
- 353. How do you configure IP restrictions dynamically with automation?
- 354. How do you configure access reviews for least privilege validation?
- 355. How do you monitor exposure of connection strings in code repos?
- 356. How do you configure bastion access patterns for SQL DB?
- 357. How do you handle compromised managed identity access?
- 358. How do you rotate managed identity credentials automatically?
- 359. How do you configure database-level firewall rules for vendors?
- 360. How do you configure tamper-proof activity logs?

Migration Edge Cases

- 361. How do you migrate databases larger than 20TB into Hyperscale?
- 362. How do you migrate SQL Server instances using transactional replication?
- 363. How do you migrate data with minimal downtime using CDC?
- 364. How do you migrate workloads dependent on cross-database queries?
- 365. How do you migrate legacy CLR code to Azure SQL DB alternatives?
- 366. How do you migrate SSIS packages to Azure Data Factory pipelines?
- 367. How do you migrate Always On AG workloads to Azure SQL DB?
- 368. How do you migrate heterogeneous workloads (Oracle/MySQL)?
- 369. How do you migrate encrypted on-prem databases to SQL DB?
- 370. How do you migrate workloads with FileStream dependencies?
- 371. How do you migrate service broker workloads into Azure?

- 372. How do you migrate jobs with complex Agent schedules?
- 373. How do you migrate linked server dependencies to Azure SQL?
- 374. How do you migrate CLR assemblies not supported in Azure SQL?
- 375. How do you migrate partitioned tables at scale?
- 376. How do you migrate SSRS reports into Power BI post-SQL migration?
- 377. How do you validate row count consistency after migration?
- 378. How do you verify LOB data migrated correctly?
- 379. How do you rollback failed migrations in DMS?
- 380. How do you perform phased migrations for SaaS tenants?

Day-to-Day Administration

- 381. How do you reset a lost admin password for Azure SQL DB?
- 382. How do you configure monitoring for all query timeouts?
- 383. How do you manage multiple environments (Dev/Test/Prod) in SQL DB?
- 384. How do you restore individual tables from point-in-time backups?
- 385. How do you refresh non-prod environments securely with masked prod data?
- 386. How do you configure alerts for schema drift?
- 387. How do you enable and test automatic plan regression correction?
- 388. How do you identify unused databases consuming cost?
- 389. How do you enforce tagging before allowing DB creation?
- 390. How do you manage database renaming in Azure SQL DB?
- 391. How do you configure alerts for dropped objects?
- 392. How do you configure versioning of database schemas?
- 393. How do you monitor tenant isolation in SaaS databases?
- 394. How do you create templates for new Azure SQL deployments?
- 395. How do you monitor and alert on transaction log space?
- 396. How do you configure usage baselines for anomaly detection?
- 397. How do you test schema changes in isolated clones?
- 398. How do you manage users without sysadmin role in SQL DB?
- 399. How do you configure SQL DB diagnostics into Event Hub?
- 400. How do you test impact of new features on existing workloads?

Cost & Optimization Advanced

- 401. How do you configure workload classifier to prioritize business workloads?
- 402. How do you analyze which queries consume the most DTUs/vCores?
- 403. How do you tune elastic pool configurations to avoid overprovisioning?
- 404. How do you schedule scaling during peak seasons automatically?
- 405. How do you enforce budget limits per team?
- 406. How do you predict monthly cost variations based on scaling events?
- 407. How do you detect zombie databases left unused for months?
- 408. How do you optimize query patterns to reduce I/O costs?
- 409. How do you configure reports for reserved instance utilization?
- 410. How do you configure storage auto-grow monitoring for cost control?
- 411. How do you configure alerts when storage reaches 80% utilization?
- 412. How do you configure performance tests before scaling up tiers?
- 413. How do you track cost impact of geo-replication?
- 414. How do you calculate break-even point between DTU vs vCore?
- 415. How do you implement governance for elastic pool over-allocation?
- 416. How do you detect inefficient use of serverless auto-pause?
- 417. How do you simulate scaling costs for different tenants?

- 418. How do you implement chargeback model per application team?
- 419. How do you detect duplicate databases consuming unnecessary costs?
- 420. How do you analyze workload fit for Hyperscale vs Business Critical?

Expert Troubleshooting & Edge Cases

- 421. How do you troubleshoot unexpected throttling despite low utilization?
 - 422. How do you detect query plan regressions post-engine upgrade?
 - 423. How do you troubleshoot automatic index recommendations failing?
 - 424. How do you troubleshoot failed query store capture?
 - 425. How do you handle tempdb contention in elastic pool environments?
 - 426. How do you troubleshoot intermittent performance degradation?
 - 427. How do you capture heap table performance regressions?
 - 428. How do you troubleshoot failed LTR backup restore?
 - 429. How do you troubleshoot query timeout in Hyperscale tier
430. How do you troubleshoot connection pooling issues in Azure SQL DB?
- 431. How do you troubleshoot slow queries when Query Store is disabled?
 - 432. How do you troubleshoot failed TDE key rotations in SQL DB?
 - 433. How do you troubleshoot login failures due to Azure AD misconfiguration?
 - 434. How do you troubleshoot replication lag in geo-replicated databases?
 - 435. How do you troubleshoot data sync errors between Azure SQL DB and on-prem?
 - 436. How do you troubleshoot memory pressure in Business Critical tier?
 - 437. How do you troubleshoot queries spilling to tempdb?
 - 438. How do you troubleshoot distributed query failures with external tables?
 - 439. How do you troubleshoot blocked system views in Azure SQL DB?
 - 440. How do you troubleshoot firewall rule conflicts across subscriptions?
 - 441. How do you troubleshoot queries failing due to MAXDOP limitations?
 - 442. How do you troubleshoot row version store space exhaustion?
 - 443. How do you troubleshoot high waits on SOS_SCHEDULER_YIELD in SQL DB?
 - 444. How do you troubleshoot missing execution plans in Query Store?
 - 445. How do you troubleshoot managed identity authentication failures?
 - 446. How do you troubleshoot TLS handshake timeouts from client apps?
 - 447. How do you troubleshoot database stuck in "Restoring" state?
 - 448. How do you troubleshoot elastic pool DTU starvation?
 - 449. How do you troubleshoot deadlocks between multiple tenants?

Disaster Recovery & Business Continuity (Advanced)

- 450. How do you test failover groups during Azure regional outage simulation?
- 451. How do you configure DR drills without impacting production?
- 452. How do you validate RTO/RPO in geo-replication scenarios?
- 453. How do you recover from failover group misconfiguration?
- 454. How do you automate DR validation across multiple databases?
- 455. How do you configure alerts for replication lag in DR setups?
- 456. How do you handle schema drift between primary and geo-secondary?
- 457. How do you restore databases to isolated subscription during DR?
- 458. How do you simulate accidental region-wide deletion recovery?
- 459. How do you configure DNS cutover validation during failover?

Automation & DevOps

- 460. How do you integrate Azure SQL DB deployments in CI/CD pipelines?
- 461. How do you automate schema drift detection with pipelines?
- 462. How do you automate scaling rules using ARM templates?
- 463. How do you use Terraform to deploy Azure SQL DB at scale?
- 464. How do you configure automated security baseline checks in DevOps?
- 465. How do you run automated performance tests after schema deployment?
- 466. How do you manage connection strings securely in DevOps pipelines?
- 467. How do you configure Blue/Green deployments for schema changes?
- 468. How do you implement automated rollback for failed deployments?
- 469. How do you automate data masking in pipeline refreshes?
- 470. How do you enforce naming conventions in Azure SQL DB deployments?

Monitoring & Observability (Advanced)

- 471. How do you configure proactive anomaly detection using Azure Monitor?
- 472. How do you integrate SQL Insights metrics with Grafana?
- 473. How do you monitor query performance across elastic pools centrally?
- 474. How do you configure alerts on unusual login patterns?
- 475. How do you monitor top resource-consuming queries at scale?
- 476. How do you configure distributed tracing from App Insights to SQL DB?
- 477. How do you baseline workload performance before scaling decisions?
- 478. How do you configure proactive alerts for deadlocks?
- 479. How do you monitor LTR backup storage costs over time?
- 480. How do you configure workload classification telemetry collection?

Hybrid & Multi-Cloud Scenarios

- 481. How do you integrate Azure SQL DB with on-prem Active Directory?
- 482. How do you replicate SQL DB data into AWS RDS for hybrid use cases?
- 483. How do you configure SQL DB as a source for multi-cloud analytics?
- 484. How do you configure data sync between Azure SQL DB and Google Cloud SQL?
- 485. How do you configure SQL DB failover strategy across clouds?
- 486. How do you secure hybrid SQL DB traffic over VPN tunnels?
- 487. How do you optimize latency in hybrid SQL DB applications?
- 488. How do you configure DNS resolution for hybrid SQL DB workloads?
- 489. How do you ensure consistent auditing across hybrid SQL environments?
- 490. How do you handle compliance across hybrid SQL workloads?

Future-Proofing & Best Practices

- 491. How do you prepare Azure SQL DB for future AI-driven workloads?
- 492. How do you configure Hyperscale tier for unpredictable growth?
- 493. How do you optimize for serverless adoption at scale?
- 494. How do you evaluate Business Critical vs Hyperscale trade-offs?
- 495. How do you configure Azure SQL DB for quantum-safe encryption readiness?
- 496. How do you enable automated index correction in long-term workloads?
- 497. How do you test new Azure SQL DB features in isolated sandboxes?
- 498. How do you build governance around SQL DB self-service provisioning?
- 499. How do you configure high availability in multi-region SaaS?
- 500. How do you enforce enterprise-wide standards for Azure SQL DB deployments?